# ESSENTIAL MATHEMATICS GRADE 10: SURFACE AREA OF SOLIDS

*CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.6 (8 Lessons)*

| SUB-STRAND OVERVIEW | |
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| **Grade Level** | 10 |
| **Subject** | Essential Mathematics |
| **Strand** | Strand 2.0: Measurements and Geometry |
| **Sub-Strand** | Sub-Strand 2.6: Surface Area of Solids |
| **Total Duration** | 8 lessons × 40 minutes = 320 minutes total |
| **Sub-Strand Content** | – Surface area of a cone — using the net of a cone to derive and apply the formula<br>– Surface area of a pyramid (square or rectangular base) — using nets to derive and apply the formula<br>– Surface area of a sphere and hemisphere — deriving and applying the formulae<br>– Surface area of a frustum of a cone — open and closed frustums<br>– Surface area of a frustum of a pyramid (square or rectangular base) — open and closed frustums<br>– Applications of surface area of cones, pyramids, spheres, hemispheres and frustums in real-life situations |
| **Learning Outcomes** | By the end of the sub-strand, the learner should be able to:<br>a) determine the surface area of cones and pyramids from the nets,<br>b) calculate the surface area of a sphere and a hemisphere,<br>c) determine the surface area of frustums of cones and pyramids,<br>d) apply the surface area of cones, pyramids and frustums in real life situations,<br>e) appreciate the application of surface area of solids. |
| **Core Competencies** | – Critical Thinking and Problem Solving: the learner uses nets of cones and pyramids to derive surface area formulae, explores different approaches to computing the surface area of open and closed frustums, and applies the knowledge to solve real-life problems such as estimating the material needed to make a bucket or lampshade.<br>– Creativity and Imagination: the learner works with peers to model frustums of cones and pyramids (both open and closed) using locally available materials such as cardboard from Nairobi market packaging, developing connection and networking skills in the process of constructing and measuring the models. |
| **Core Values** | – Social Justice: the learner accords equal opportunities to all group members when discussing the uses of pyramids and frustums in day-to-day life, ensuring every voice is heard during group work.<br>– Responsibility: the learner observes safety precautions and takes care of peers while using cutting tools and locally available materials to model frustums of cones and pyramids, and properly returns shared instruments after use. |
| **Science & Engineering Practices** | – Developing and Using Models: learners construct physical nets and 3-D models of cones, pyramids, and frustums using cardboard, then use the unfolded nets to derive and verify surface area formulae — connecting the 2-D representation to the 3-D solid.<br>– Using Mathematics and Computational Thinking: learners apply geometric formulae systematically, substitute measured |

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| | dimensions into expressions for slant height, base area and lateral surface area, and check reasonableness of calculated surface areas against physical estimates of material quantities.<br>– Obtaining, Evaluating and Communicating Information: learners gather measurements from physical objects in their environment (buckets, lampshades, grain silos at Eldoret farms), evaluate which formula applies, and communicate their findings and solution strategies to peers through presentations and model displays. |
| **Pertinent & Contemporary Issues (PCIs)** | – Environment and Technology: the learner collects and uses locally available materials (cardboard, old packaging, newspaper) to model frustums of cones and pyramids for use as learning resources, reinforcing the principle of resourcefulness and environmental responsibility. |
| **Career Connections** | – Manufacturing and Packaging Industry: knowledge of surface area of cones, pyramids and frustums is essential for calculating the amount of sheet metal or packaging material needed to produce containers such as the conical paper cups used in Nairobi hospitals or grain storage bins in Rift Valley farms.<br>– Architecture and Construction: computing surface area of pyramidal roofs and domed structures (e.g., the conical roofs of traditional Kikuyu huts or the hemispherical domes on mosques in Mombasa) is critical for estimating roofing materials and costs.<br>– Hospitality Industry: chefs and caterers in Kenya's hotel industry use knowledge of the surface area of cones and hemispheres when designing moulded food presentations, portion sizes, and decorative cake coverings.<br>– Sports Management: surface area calculations for spherical balls (soccer, basketball) are used in manufacturing, quality control, and equipment budgeting for sports clubs and school teams across Kenya. |
| **Focus for Lessons** | This unit develops learners' ability to calculate the surface area of cones, pyramids, spheres, hemispheres, and frustums by first exploring the 2-D nets of each solid, then deriving and applying formulae to real-life objects found in the Kenyan environment. A hands-on, model-building approach is used throughout, anchored in the puzzling observation that everyday objects — from water buckets and grain silos in the Rift Valley to lampshades and conical paper cups — have specific shapes whose outer covering can be precisely quantified. Learners work collaboratively to construct, measure, and compute, progressively building from simple solids to composite frustums over the eight lessons. |
| **Driving Question / Key Inquiry Questions** | DRIVING QUESTION: How do we calculate exactly how much material is needed to cover or construct any solid shape — and why does the shape of a container matter even when the volume stays the same?<br><br>KICD KEY INQUIRY QUESTION:<br>1. How are solids used in real life? |
| **Anchoring Phenomenon** | A hardware shop along Kirinyaga Road in Nairobi sells three different metal buckets: a straight-sided cylindrical bucket, a frustum-shaped (tapered) bucket wider at the top than the bottom, and a conical bucket. All three buckets hold exactly 10 litres of water. The shop owner tells the class that the tapered (frustum-shaped) bucket is the most expensive because "it uses the most metal." Students are puzzled — if all three buckets hold the same amount of water, why would the shape affect how much metal sheet is needed to make it? When learners handle physical models of the three buckets and trace their unfolded nets onto paper, they notice that the frustum's net is unexpectedly large and complex compared to the cylinder. This creates genuine curiosity: how do we calculate exactly how much sheet metal is needed for each shape, and is the shopkeeper correct that the frustum uses the most material? This anchoring phenomenon drives the entire unit as learners develop tools to compute the surface area of each type of solid. |
| **Storyline Thread** | Lesson 1 — Anchoring Phenomenon & Nets of Cones: Learners encounter the bucket phenomenon (cylindrical vs. frustum vs. conical buckets holding the same 10 L) and make initial predictions about which uses the most metal. They then construct and unfold the net of a cone from cardboard, identify the sector (lateral surface) and circular base, and use measurements of the net to |

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| | derive the formula: Total Surface Area of a cone = $\pi rl + \pi r^2$, where l is the slant height. Learners compute the surface area of a conical paper cup similar to those used at Nairobi hospital water dispensers.<br><br>Lesson 2 — Surface Area of Pyramids (Square and Rectangular Base): Learners construct nets of square-based and rectangular-based pyramids using cardboard (resembling the pyramid-shaped gift boxes from the phenomenon). They identify the base and four triangular faces, derive the formula for total surface area, and apply it to calculate the material needed for a pyramid-shaped gift box with given dimensions. Learners compare their computed areas with physical estimates made by measuring the unfolded net.<br><br>Lesson 3 — Surface Area of a Sphere: Learners use spherical objects (soccer balls, oranges, marbles) to estimate the diameter, then explore the relationship between the surface area of a sphere and four times the area of its great circle (great circle demonstration using orange peel flattening activity). Learners derive and apply the formula SA = $4\pi r^2$ and solve problems involving spherical objects such as painting a dome structure similar to those seen in Mombasa mosques.<br><br>Lesson 4 — Surface Area of a Hemisphere: Building on Lesson 3, learners slice a spherical model in half and calculate the surface area of a hemisphere (curved surface = $2\pi r^2$; total including flat circular face = $3\pi r^2$). Learners apply the formula to problems such as estimating the canvas needed to cover a hemispherical tent used at a Maasai Mara safari camp. Mid-unit formative check — learners revisit the DQB and update their initial predictions about the three buckets.<br><br>Lesson 5 — Introduction to Frustums: Nets and Vocabulary: Learners physically cut a conical model along a plane parallel to the base to produce a frustum, observing the two circular ends (radii R and r) and the slant height l. They unfold the frustum net and identify it as the difference between two sectors. Learners are introduced to the formula for the curved surface area of a frustum of a cone: $\pi(R + r)l$, and compute l from the Pythagorean relationship $l = \sqrt{[h^2 + (R - r)^2]}$.<br><br>Lesson 6 — Total Surface Area of Frustums of Cones (Open and Closed): Learners calculate the total surface area of closed frustums (adding both circular ends: $\pi R^2 + \pi r^2$) and open frustums (adding only the bottom or neither circular face), directly connecting to the tapered metal bucket in the anchoring phenomenon. Small groups measure physical frustum-shaped containers (buckets, plastic tumblers) brought from home, compute predicted surface areas, then compare with manufacturer specifications. Learners resolve the anchoring question: does the frustum bucket indeed use more metal than the cylinder or cone?<br><br>Lesson 7 — Surface Area of Frustums of Pyramids: Learners extend frustum concepts to pyramidal frustums (square or rectangular base), identifying the four trapezoidal lateral faces and the two rectangular/square bases. They derive the formula by summing the areas of all faces, apply it to real-life objects such as a truncated pyramid-shaped planter box or the base of a trophy stand. Learners build a physical model of a pyramidal frustum from cardboard and verify computed surface area against measurements of the unfolded net.<br><br>Lesson 8 — Consolidation, Applications and Model Revision: Learners revisit the anchoring phenomenon with full computational evidence, comparing the surface areas of all three bucket types (cone, frustum, cylinder) of equal volume. Groups present their solutions and models to the class, justifying the shopkeeper's claim using mathematical evidence. Learners complete a portfolio task: selecting one real-life Kenyan object (e.g., a conical grain store, a frustum water tank, a spherical water tank in Kisumu), computing its surface area, and recording the steps. The DQB is finalised, and the cumulative surface area model is updated to include all solids studied. |
| **Supporting Phenomena** | – A mandazi (Kenyan doughnut) vendor in Kisumu wraps conical paper cups with a single piece of paper — learners observe |

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| | the cone being unrolled flat and wonder how the vendor cuts the paper to exactly the right size with no waste.<br>– A football manufacturer in Nakuru prints a pattern of hexagonal and pentagonal panels on flat leather before stitching them into a sphere — learners wonder how the total flat area of the panels relates to the surface area of the finished spherical ball.<br>– A farmer in the Rift Valley builds a conical thatched roof over a circular granary — learners observe that the thatch material required depends on the slant height and base circumference, raising the question of how these measurements translate into a total area to be covered.<br>– A pyramid-shaped gift box sold in Westgate Shopping Mall in Nairobi is made from a single flat cardboard net — learners unfold the box and are surprised by the cross-shaped net, prompting questions about how the four triangular faces and square base contribute to the total surface area. |

| LESSON 1 (80 minutes): Anchoring Phenomenon & Nets of Cones: How Much Metal Makes a Bucket? | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | To introduce the anchoring phenomenon of three equal-volume metal buckets (cylindrical, frustum, and conical) sold along Kirinyaga Road, Nairobi, and to derive the total surface area formula for a cone by constructing and unfolding its net. |
| **Knowledge** | Learners will know that: (i) a cone is composed of a circular base and a curved lateral surface; (ii) when unfolded, the lateral surface of a cone forms a sector of a circle with radius equal to the slant height l; (iii) the total surface area of a cone is $TSA = \pi rl + \pi r^2$, where r is the base radius and l is the slant height; (iv) the slant height l relates to the perpendicular height h and base radius r by the Pythagorean theorem: $l = \sqrt{(r^2 + h^2)}$. |
| **Skills** | Learners will be able to: (i) construct a physical net of a cone from cardboard and identify its component parts; (ii) measure the radius and arc length of the sector from the net to derive the lateral surface area formula; (iii) calculate the total surface area of a cone given its radius and slant height or perpendicular height; (iv) apply the TSA formula to a real-world conical object (a paper cup used at Nairobi hospital water dispensers). |
| **Attitudes** | Learners will appreciate that: (i) the shape of a container matters even when its volume is fixed; (ii) mathematical formulae are tools derived from physical exploration, not arbitrary rules; (iii) collaborative investigation and peer discussion enrich mathematical understanding. |
| **Key Inquiry Question** | How are solids used in real life? |
| **Purpose in Storyline** | This is the opening lesson of the unit. It anchors every subsequent lesson in the driving question by confronting learners with a genuine puzzle — three buckets, same 10-litre volume, different amounts of metal — and begins building the mathematical tools needed to resolve it. By the end of this lesson learners can compute the surface area of the conical bucket, giving them the first data point in the unit-long comparison. The DQB opened today will grow with each lesson until learners can definitively answer whether the shopkeeper's claim is correct. |
| **Safety Notes** | Learners use scissors and craft knives to cut cardboard nets. Remind learners to always cut away from the body and keep fingers clear of the blade path. Collect all cutting tools after the activity. Ensure desks are clear of bags and loose items before cutting begins. |

## B. LESSON OVERVIEW

Learners open the unit by encountering three physical model buckets (cylindrical, frustum-shaped, and conical) — all holding exactly 10 litres — displayed at the front of the class as props representing a hardware shop along Kirinyaga Road, Nairobi. After recording initial predictions about which bucket uses the most sheet metal, learners construct a cone net from cardboard, unfold it, and measure its sector and circular base to derive $TSA = \pi rl + \pi r^2$. They then apply the formula to compute the surface area of a conical paper cup (like those at Nairobi hospital water dispensers, $r = 3.5$ cm, $l = 8$ cm). The lesson closes with the opening of the Driving Question Board (DQB) where learners post questions and partial insights that will be resolved across the unit.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Phase 1 — Predict**<br>📹 **VIDEO:** [Writing proportional equations from tables]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | Learners observe three labelled model buckets placed on the teacher's desk: Bucket A (cylinder), Bucket B (frustum — wider at top), and Bucket C (cone). The teacher states: 'A hardware shopkeeper on Kirinyaga Road in Nairobi tells a customer that all three buckets hold exactly 10 litres of water — but Bucket B costs the most because it uses the most metal sheet to make.' Learners individually write in their exercise books: (a) which bucket they predict uses the most metal sheet and why; (b) which uses the least metal sheet and why; (c) one question they have about how to find out. Learners then share predictions in pairs (Think-Pair-Share) before 4 cold-call volunteers share with the class. Predictions are NOT confirmed or denied at this stage — they are posted on the Prediction Wall (a section of the blackboard reserved for this purpose). | 1. Place the three labelled bucket models prominently on the desk before learners enter. As class settles, gesture to the buckets: 'Before I say a word about maths, I want you to just look at these three buckets. What do you notice?' Allow 15 seconds of silent observation. 2. State the phenomenon clearly and slowly: 'The shopkeeper on Kirinyaga Road says Bucket B — the tapered one — is the most expensive because it uses the most metal. All three hold exactly 10 litres. Do you believe the shopkeeper?' 3. Instruct: 'Open your exercise books. You have 4 minutes. Write your prediction — which bucket uses the most metal, which uses the least, and one question you have. Work alone — no discussion yet.' Set a visible 4-minute timer. 4. After 4 minutes: 'Turn to your partner — 90 seconds — share your prediction and your question.' After 90 seconds, call TIME. 5. Cold-call 4 learners (aim for gender balance and mixed confidence levels): 'Aisha, tell me your prediction and your reason.' After each response, say 'Thank you — I'm not telling you if you are right yet. That is exactly what we are going to find out.' 6. Write each prediction on the Prediction Wall with the learner's name: e.g., 'Wanjiru: Cylinder uses most metal because it is tallest.' 7. Pose the focus question for the lesson: 'To answer this, we first need a way to calculate the surface area of each shape. We start today with the cone — Bucket C. Let us figure out where the formula comes from.' | Think-Pair-Share with a Prediction Wall. This strategy activates prior knowledge about 3D shapes and volume/surface area, reveals misconceptions (e.g., 'more volume = more surface area'), and creates genuine cognitive dissonance — the phenomenon is puzzling enough that learners are motivated to find the answer. Posting predictions publicly means learners will return to revise them, reinforcing metacognitive growth across the unit. | Circulate during individual writing time and read 4–5 books. Look for: (i) whether learners attempt to use any quantitative reasoning or rely purely on visual intuition; (ii) whether learners distinguish between volume and surface area in their language. FLAG: if more than half of the class cannot write any reason (only a guess), pause and ask 'What does it mean to use more metal sheet?' to surface the concept of surface area before proceeding. |
| **Phase 2 —** | In groups of 3–4, learners receive | 1. Distribute net kits to groups. | Concrete-Pictorial-Abstract (CPA) | Collect data from circulating visits: |

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| **Observe**<br>📹 **VIDEO:** [Writing proportional equations from tables]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | a pre-scored cardboard net kit. The kit contains: (i) a pre-cut sector (the lateral face of a cone, radius = 10 cm, arc length ≈ 22 cm) and (ii) a pre-cut circle (radius = 3.5 cm, representing the base). Groups are also given a ruler, a piece of string, and a marker. Task card instructions: Step 1 — Using the string, measure the arc length of the sector and record it. Step 2 — Measure the straight edge (radius) of the sector and record it — this is the slant height l. Step 3 — Measure the radius r of the circular base. Step 4 — Roll the sector into a cone and attach the base using a small piece of tape. Observe how the arc of the sector becomes the circumference of the base circle. Step 5 — Unroll it and lay it flat again. Record: 'The arc length of the sector = the circumference of the base circle = 2πr.' Step 6 — Sketch the net in your exercise book, labelling l (slant height), r (base radius), and the sector arc. Learners record all measurements in a table drawn in their exercise books: \| Part \| Shape \| Measurement \|. A short 3-minute teacher-narrated demonstration video (pre-loaded on ARES offline) showing a cone being unrolled in slow motion is available for groups that finish early or need visual support. | Say: 'You have 15 minutes. Your job is not yet to find a formula — your job is to observe carefully and record what you see. The task card will guide you step by step.' 2. Circulate to each group at least twice. At the first pass (minutes 2–5), check that learners are measuring the arc with string correctly. If a group is struggling, ask: 'What shape is this curved edge? How would you measure the length of a curve?' — wait 10 seconds for a response before hinting to use the string. 3. At the second pass (minutes 8–12), prompt groups that have rolled the cone: 'Look at where the arc meets the base circle. What do you notice about those two lengths?' Wait 10 seconds. Listen for 'they are the same' or 'the arc fits around the circle.' 4. At minute 13, give a 2-minute warning. 5. Bring class back together. Cold-call 2 groups to report their arc length measurement and base circumference: 'Group Three — Kamau, what did you measure for the arc length, and what did you get for 2πr of the base? Are they the same?' 6. Write on the board: Arc length of sector = Circumference of base = 2πr. Circle this and say: 'This is the key observation we will use in the next phase to build the formula.' | with guided observation. Learners handle the physical net before seeing any formula, so when the formula is derived in Phase 3 it is anchored to their tactile experience. The step-by-step task card ensures all groups gather the same critical data point (arc = 2πr) regardless of pace. The act of rolling and unrolling the net physically embeds the relationship between the sector and the cone in procedural and conceptual memory simultaneously. | (i) Can learners correctly identify the slant height l as the straight radius of the sector? (ii) Do learners record arc length ≈ base circumference (within ±0.5 cm measurement error)? (iii) Can learners label the sketch with correct variable names? FLAG: if learners call the sector radius 'the height of the cone', intervene with the physical model — point to the perpendicular height inside the rolled cone vs. the slant edge. This is a critical misconception to resolve before Phase 3. |
| **Phase 3 — Explain**<br>📹 **VIDEO:** [Writing proportional equations from tables]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES | Whole-class guided derivation followed by individual practice. The teacher leads a step-by-step derivation of the cone TSA formula on the board, with learners completing a structured note-frame in their exercise books (pre-printed or copied): 'The lateral surface of a cone is a sector of a circle with radius = ___ (slant height l) and | 1. Begin at the board: 'Let us use what your groups just observed to build the formula together. Do not copy yet — just watch and think.' Draw the unrolled net (sector + circle) on the board with labels l, r, arc. 2. Step 1 — Ask: 'What fraction of a full circle is this sector?' Pause 12 seconds. Cold-call: 'Fatuma, how do we express | Guided Derivation with Structured Note-Frame. Rather than presenting the formula as a given fact, the teacher guides learners through each logical step, and learners fill in a partially completed frame. This strategy ensures that the formula is understood as a consequence of geometry rather than memorised. The immediate | Circulate during the 6-minute individual work time. Specific indicators: (i) Does the learner correctly substitute r = 3.5 and l = 8 (not h)? (ii) Does the learner compute πrl and πr² separately before adding? (iii) Is the final answer 126.5 cm²? FLAG: if more than 6 learners substitute the wrong value for l, stop the class |

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| for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | arc length = ___. Area of a full circle with radius l = ___. The sector is a fraction of this full circle. The fraction = (arc length) ÷ (full circumference) = $2\pi r \div 2\pi l$ = ___. Therefore, Lateral Surface Area = $(r/l) \times \pi l^2$ = ___. Adding the base: TSA = $\pi rl + \pi r^2$.' After the derivation, the teacher connects back to the phenomenon: 'So for Bucket C — the conical bucket — we now have a tool to find the metal sheet area.' Learners then individually solve the application problem: 'A conical paper cup used at a water dispenser at Kenyatta National Hospital has base radius r = 3.5 cm and slant height l = 8 cm. Calculate the total surface area of the cup. (Use $\pi = 22/7$.)' Early finishers solve an extension: 'The perpendicular height of the conical bucket (Bucket C) is 28 cm and its base radius is 10.5 cm. Find the slant height and then calculate the total surface area of the bucket.' | that fraction using arc length?' Write: fraction = arc / $(2\pi l) = 2\pi r / (2\pi l) = r/l$. 3. Step 2 — Ask: 'What is the area of a full circle with radius l?' Pause 10 seconds. Cold-call: 'Otieno, what is the area?' Write: $\pi l^2$. 4. Step 3 — 'So the sector area = $(r/l) \times \pi l^2 = \pi rl$. Say that with me — pi, r, l.' Class repeats. 5. Step 4 — 'The base is a circle with radius r. Its area is?' Cold-call: 'Mwangi?' Write $\pi r^2$. 6. Step 5 — 'Therefore Total Surface Area = $\pi rl + \pi r^2$. Write it in your note frame. Put a box around it.' 7. Connect to phenomenon: 'We now have the tool for Bucket C. Let us use it.' 8. Read the KNH paper cup problem aloud. Say: 'Work alone — 6 minutes. Show every step.' Set timer. 9. After 6 minutes, cold-call one learner to write their solution on the board. Class checks. Correct answer: Lateral SA = $(22/7) \times 3.5 \times 8 = 88$ cm²; Base = $(22/7) \times 3.5^2 = 38.5$ cm²; TSA = 126.5 cm². 10. Pose extension to fast finishers. After whole-class check, explicitly say: 'Notice — the cone formula uses slant height l, not perpendicular height h. This is a common error. Always check which height you are given.' | application to the KNH paper cup keeps the abstract derivation tethered to the real world and to the unit phenomenon. | and re-display the labelled net diagram: 'Point to l on your net sketch. That is what goes into the formula — not the height inside the cone.' |
| **Phase 4 — Driving Question Board (DQB) Creation**<br>📹 **VIDEO:** Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES | Each learner receives two sticky notes (or small torn pieces of paper if sticky notes are unavailable). On the FIRST note, learners write one thing they are NOW wondering about the bucket phenomenon — a genuine question that this lesson has not yet answered. On the SECOND note, they write one thing they now know for certain that they did not know at the start of the lesson. Learners stick or tape their notes | 1. Distribute two sticky notes per learner. Say: 'Think quietly for 2 minutes. On the PINK note, write one question you still have about the buckets or about surface area. On the YELLOW note, write one thing you are now certain about that you were not certain about when you walked in.' 2. After 2 minutes, invite learners to come up row by row and place their notes on the DQB. 3. Read 5–6 questions aloud with genuine | Driving Question Board (DQB) — a visible, cumulative record of the class's growing understanding. The DQB externalises metacognition: learners can see what they collectively know and don't know. Clustering questions previews the unit structure for learners without the teacher lecturing about it. Connecting back to the Prediction Wall creates continuity and reminds learners that their initial intuitions are part of | Read the 'We Now Know' notes quickly as they are posted. Indicators: (i) Do learners correctly state the cone TSA formula in their own words? (ii) Do any notes reveal persistent confusion (e.g., 'I now know that surface area is the same as volume')? FLAG: Collect and read all notes after class. Use misconceptions identified in the 'We Now Know' column to design the opening review activity of Lesson 2. |

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| for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | onto a dedicated section of the blackboard (or a large sheet of manila paper) divided into two columns: 'We Are Wondering…' and 'We Now Know…'. The teacher reads 5–6 questions aloud and organises similar questions into clusters with a chalk circle. The teacher labels emerging clusters: e.g., 'Cylinder surface area', 'Frustum surface area', 'Comparing all three', 'What is slant height for the frustum?'. The teacher then says: 'Every lesson in this unit will add to the NOW KNOW column and help us answer questions in the WONDERING column. By the end of the unit, we will be able to tell the shopkeeper on Kirinyaga Road whether he is correct.' | curiosity: 'Listen to this one — Akinyi asks: If the frustum is like a cone with the top cut off, do we just subtract the top cone's area? Interesting! Let us park that — we will answer it in Lesson 3.' 4. Cluster visibly: physically draw chalk circles around related notes. 5. Point to the Prediction Wall: 'Your predictions are still up there. We have not changed them yet. We have only answered one part of the puzzle — the cone. We have four more shapes to go.' 6. Close with: 'The DQB stays on the wall for the whole unit. Every lesson, we will return to it.' | the learning process, not errors to be hidden. | |
| **Phase 5 — Model Building**<br>📹 **VIDEO:** Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Learners draw the first entry in their unit Model Journal — a dedicated page (or folded A4 sheet kept in their exercise book) titled 'My Growing Model: How Much Metal Makes a Bucket?' At this stage, learners draw and annotate only what they have confirmed so far: a diagram of the conical bucket (Bucket C) with its net beside it, labelling r, l, the sector (lateral surface), and the circular base. Beneath the diagram, they write the confirmed formula: TSA of Cone = $\pi rl + \pi r^2$ and their calculated value for the KNH paper cup (TSA = 126.5 $cm^2$). They leave space labelled 'Cylinder (Lesson 2)', 'Sphere/Hemisphere (Lesson 2/3)', and 'Frustum (Lesson 3/4)' as placeholders for future lessons. Learners who finish early annotate their model with a note explaining WHY the arc of the sector equals $2\pi r$, in their own words. | 1. Say: 'In the last 8 minutes, you are going to start your Model Journal for this unit. This is not a neat final answer — it is a growing record of your thinking. It will look incomplete today — that is correct and expected.' 2. Show a teacher exemplar Model Journal page (hand-drawn on the board or on a prepared A3 poster) with the cone section filled in and the other sections blank with placeholder labels. 3. Say: 'Copy the structure. Fill in what we confirmed today. Leave the rest blank — you will fill it in as we go.' 4. Circulate and check that: (i) net diagram includes both sector and base with correct labels; (ii) formula is written correctly with variables defined; (iii) placeholders for other shapes are present. 5. With 2 minutes remaining, say: 'Pause where you are. We will return to this in every lesson. By Lesson 5, this page will be full, and you will be able to | Cumulative Model Journal with deliberate placeholders. The incomplete model is a pedagogical tool — the blank sections create productive tension and a sense of an ongoing investigation. The physical act of drawing, labelling, and writing the formula consolidates the Phase 3 derivation in a personally authored artefact. Returning to the same page in future lessons means learners can literally see their understanding growing, reinforcing self-efficacy. | Collect 6–8 Model Journals (or photograph them) to review after class. Indicators: (i) Is the net diagram correctly labelled with r and l (not r and h)? (ii) Is the formula written as TSA = $\pi rl + \pi r^2$ (not TSA = $\pi r^2 + \pi rh$)? (iii) Are placeholders present for future shapes? Use this as the basis for targeted feedback returned at the start of Lesson 2. FLAG: Any learner who writes $\pi rh$ instead of $\pi rl$ needs individual follow-up at the start of Lesson 2. |

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| | | answer the shopkeeper's question.' 6. Close the lesson by returning to the phenomenon: 'Today we confirmed the surface area of one bucket — the conical one, Bucket C. The question of whether the shopkeeper is right is still open. What do we need to find out next?' Take 2–3 verbal responses. Expected: 'We need to find the surface area of the cylinder and the frustum.' | | |

## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the phenomenon create genuine curiosity, or did learners simply wait for the teacher to tell them the answer? If the latter, consider bringing a real 10-litre plastic container and an actual hardware price tag to Lesson 2 to deepen the authenticity of the context. (2) During the net construction activity, did all groups correctly identify the arc length as equal to $2\pi r$? If two or more groups did not, open Lesson 2 with a 5-minute re-demonstration using the physical model before moving to the cylinder. (3) Were learners consistently confusing slant height $l$ with perpendicular height $h$? If so, create a dedicated 'l vs. h' anchor chart to display permanently for the rest of the unit. (4) Review the DQB 'We Are Wondering' notes tonight — do learners' questions map onto the lessons planned? If learners have raised questions about pyramids or frustums ahead of schedule, acknowledge these on the DQB and clarify the unit sequence tomorrow. (5) Check the Model Journal samples — are learners treating this as a 'neat copy' or as a genuine thinking tool? Praise rough, annotated journals explicitly in Lesson 2 to reshape this norm if needed.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| **What did I observe?** | Three metal bucket models (cylindrical, frustum-shaped, conical) all holding 10 litres; a cardboard net of a cone (sector + circle base) which learners assembled and unrolled; the key observation that the arc length of the sector equals the circumference of the cone's base (arc = $2\pi r$). |
| **What did I learn?** | The total surface area of a cone = $\pi rl + \pi r^2$, where $r$ is the base radius and $l$ is the slant height; the lateral surface of a cone unfolds into a sector of a circle with radius $l$; the slant height $l \neq$ perpendicular height $h$; for the KNH paper cup ($r = 3.5$ cm, $l = 8$ cm), TSA = 126.5 $cm^2$. |
| **How does this explain the phenomenon?** | The formula TSA = $\pi rl + \pi r^2$ is derived by expressing the sector as a fraction ($r/l$) of a full circle of radius $l$, giving lateral area = $\pi rl$, then adding the base area $\pi r^2$; the shape of a container affects its surface area even when the volume is fixed — this is why the shopkeeper's claim about the frustum bucket may be mathematically justified, and we will verify it fully by computing and comparing all three buckets' surface areas across the unit. |

| LESSON 2 (80 minutes (2 × 40-minute lessons)): Surface Area of Pyramids (Square and Rectangular Base) | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Learners derive and apply the formula for the total surface area of square-based and rectangular-based pyramids by constructing and measuring nets, connecting their calculations to the real-world question of how much sheet metal or cardboard is needed to construct a pyramid-shaped object. |
| **Knowledge** | Learners will know: (a) the names and properties of the faces of a square-based and rectangular-based pyramid (one square/rectangular base and four congruent or paired triangular faces); (b) the meaning of slant height and how it differs from the vertical height; (c) the formula for the total surface area of a square-based pyramid: $TSA = l^2 + 2ls$, where $l$ is the base side length and $s$ is the slant height; (d) the formula for a rectangular-based pyramid: $TSA = lb + l\sqrt{h^2 + (b/2)^2} + b\sqrt{h^2 + (l/2)^2}$ (derived from summing the base and four triangular face areas); (e) how to compute the area of each triangular face using $A = \frac{1}{2} \times \text{base} \times \text{slant height}$. |
| **Skills** | Learners will be able to: (a) construct an accurate net of a square-based and rectangular-based pyramid from cardboard; (b) identify and measure the base dimensions and slant height from the net; (c) calculate the area of each triangular face and the base; (d) sum component areas to determine total surface area; (e) compare computed surface area with a physical estimate obtained by measuring the unfolded net; (f) apply the formula to find the amount of material needed for a pyramid-shaped gift box with given dimensions. |
| **Attitudes** | Learners will: (a) appreciate the real-life relevance of surface area calculations in packaging, construction, and manufacturing; (b) develop patience and precision in measurement and net construction; (c) show collaborative spirit by sharing tools and checking each other's measurements. |
| **Key Inquiry Question** | How are solids used in real life? |
| **Purpose in Storyline** | In Lesson 1 learners explored the surface area of cones. Lesson 2 extends the evidence-gathering sequence to pyramids — a second solid from the hardware-shop phenomenon. By constructing nets of pyramid-shaped gift boxes and computing their surface areas, learners add a new data point to their comparison of 'how much material each shape requires,' directly advancing the driving question. Lesson 3 will address spheres and hemispheres before the class turns to frustums in later lessons. |
| **Safety Notes** | Learners use scissors and craft knives to cut cardboard nets. Teacher must remind learners to cut away from the body and to keep fingers clear of the blade path. Craft knives, if used, must be handled only by the teacher or under direct supervision. Rulers should not be used as cutting guides with craft knives by learners. Scrap cardboard pieces must be collected and disposed of safely to avoid tripping hazards. |

## B. LESSON OVERVIEW

Learners construct physical nets of square-based and rectangular-based pyramids from cardboard, mirroring the pyramid-shaped gift boxes sold near the hardware shop on Kirinyaga Road from the anchoring phenomenon. They identify the base and four triangular faces on the net, measure the base dimensions and slant height, and derive the total surface area formula by summing component areas. They then calculate the surface area of a specific pyramid-shaped gift box, compare their formula-based result with a physical estimate made by tracing and measuring the unfolded net, and record findings on their Driving Question Board. The lesson builds directly on Lesson 1 (surface area of cones) and sets up the forthcoming comparison of material use across the cylinder, frustum, and pyramid shapes central to the phenomenon.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Phase 1 — Predict**<br>📹 **VIDEO:** [Volume of a Pyramid - Overview]<br>Source: CK-12<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | Each learner receives a small pre-cut pyramid-shaped cardboard gift box (or a photograph of one, referencing the gift-box shop near Kirinyaga Road). Before unfolding or measuring anything, learners individually write a prediction in their exercise books: 'If I unfolded this pyramid completely flat, sketch what you think the net would look like. How many separate faces would you see? Which face do you think has the largest area?' Learners then share predictions with a shoulder partner (Think-Pair-Share, 2 minutes). Selected pairs share with the class. | Distribute one pyramid box (or printed photograph) per learner pair. Say: 'Before we open or measure anything, I want you to think like an engineer at a cardboard factory in Thika — you need to know how much flat cardboard sheet to order before you fold a single box. In your book, sketch what you think this pyramid looks like when fully unfolded.' Allow 3 minutes of individual silent thinking. [WAIT TIME: 15 seconds after posing the question before any learner speaks.] After Think-Pair-Share, cold-call 3 pairs: ask 'What did you sketch? How many faces did you predict?' Record diverse predictions on the chalkboard under the heading 'Our Predictions.' Do NOT correct yet. Ask: 'Do you think a pyramid-shaped gift box uses more or less cardboard than a cone-shaped box of similar size — which we studied last lesson?' [WAIT TIME: 10 seconds.] Cold-call 2 learners. Record responses next to Lesson 1 cone findings already on the Driving Question Board. | Think-Pair-Share with public prediction recording. This activates prior knowledge from Lesson 1 (cone nets) and creates a visible record of misconceptions (e.g., learners often predict only 4 faces, forgetting the base, or confuse slant height with vertical height). The public record creates cognitive dissonance that the Observe phase will resolve. | Observable indicator: Learner sketches show at least an attempt to draw a 2D unfolded shape with distinct triangular regions. Listen for misconceptions: (a) learners who draw only 4 triangles with no base — flag these learners for targeted questioning during the Observe phase; (b) learners who confuse the pyramid with a cone. Teacher circulates and notes 2–3 sketches to display anonymously during the Explain phase. |
| **Phase 2 — Observe**<br>📹 **VIDEO:** [Volume of a Pyramid - Overview]<br>Source: CK-12<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar | Groups of 3–4 learners receive a pre-scored (but not yet cut) cardboard net template of a square-based pyramid (base side = 10 cm, slant height = 13 cm) and a second template for a rectangular-based pyramid (base 8 cm × 6 cm, slant height of longer triangular faces = 12 cm, slant height of shorter triangular faces = 12.6 cm). Groups: (1) carefully cut out the net; (2) label every face (Base, Face A, Face B, Face C, Face D) before folding; (3) measure and record every | Before cutting begins, hold up the scored net and say: 'This flat shape is called the net of the pyramid — it is exactly what a sheet-metal worker in Kamukunji jua kali workshop would cut from a flat metal sheet before hammering it into shape. Every centimetre of this net will become part of the bucket or box.' Demonstrate labelling one face: 'I am calling this triangle Face A. In your groups, label all faces before you cut — this is non-negotiable because we need to track every face when we | Constructing and labelling physical nets (Concrete–Representational–Abstract progression). By physically cutting, folding, and colour-coding each face, learners build a concrete referent for every term in the formula. The measurement table provides a structured representational bridge between the physical net and the algebraic formula derived in the Explain phase. The rectangular-base net deliberately introduces asymmetry, forcing learners to see that the formula must account for | Observable indicators during circulation: (a) All five faces are labelled correctly before cutting — if not, redirect before cutting proceeds. (b) The measurement table is completed with correct units (cm and cm²). (c) Learners can point to 'slant height' on the net when asked — if a learner points to the vertical height of the 3D pyramid, this is a critical misconception to address whole-class before the Explain phase. Teacher uses a quick show-of-hands 'traffic light': 'Thumbs up if |

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| readings | dimension with a ruler; (4) fold the net into the 3D pyramid and verify it closes correctly; (5) unfold again and, using a pencil, shade each triangular face a different colour to distinguish them. Groups record all measurements in a table: Face \| Shape \| Base of triangle (cm) \| Slant height (cm) \| Area ($cm^2$). The rectangular-based pyramid step surfaces the key insight that opposite triangular faces are congruent pairs but the two pairs differ from each other. | calculate area.' [WAIT TIME: 10 seconds after each instruction to allow groups to act.] Circulate during cutting; at the 8-minute mark, pause the class: 'Hands up — how many groups have found that they have exactly FIVE faces on the net?' [Cold-call 1 group that did NOT raise their hand: 'Tell me what you have found — how many faces do you count?'] After folding, ask: 'When you folded the net, did ANY face overlap another? What does that tell us about how the net was designed?' [WAIT TIME: 12 seconds.] Cold-call 2 groups. After the rectangular-base net: 'Look at the four triangular faces of the rectangular pyramid. Are all four triangles identical? Discuss with your group for 1 minute, then I will ask three groups to report.' [WAIT TIME after 1 minute — cold-call 3 groups.] | two pairs of non-identical triangles. | you can see the slant height on your unfolded net right now; thumbs sideways if unsure.' Address all thumbs-sideways learners immediately. |
| **Phase 3 — Explain**<br>📹 **VIDEO:** Volume of a Pyramid - Overview<br>Source: CK-12<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Whole-class guided derivation. Learners use their completed measurement tables to answer: 'What is the total area of all the cardboard in your square-pyramid net?' They sum the areas: Base area + 4 × (area of one triangular face). The teacher guides them to express this algebraically as TSA = $l^2 + 4 \times (\frac{1}{2} \times l \times s) = l^2 + 2ls$. Learners then verify numerically: TSA = $10^2 + 2(10)(13) = 100 + 260 = 360$ $cm^2$. They compare this with the physical estimate (measuring the total area of the unfolded net by counting grid squares if graph-paper backing is used, or by multiplying measured dimensions). For the rectangular pyramid, learners derive TSA = $lb + 2 \times (\frac{1}{2} \times l \times s_1) + 2 \times (\frac{1}{2} \times b \times s_2) = lb + ls_1 + bs_2$. Application problem: 'A gift shop on Biashara Street in | Begin by projecting (or drawing) a large net of the square pyramid on the board with the five faces clearly labelled. Say: 'Let's build the formula the same way a jua kali artisan would price a job — face by face.' Point to the base: 'What is the shape of the base? What is its area?' [WAIT TIME: 10 seconds. Cold-call 1 learner.] Write: Base = $l^2$. Point to one triangular face: 'What is the area of this triangle in terms of the base side l and the slant height s?' [WAIT TIME: 12 seconds. Cold-call 1 learner who has NOT yet spoken.] Write: One triangle = $\frac{1}{2} \times l \times s$. 'How many such triangles does a square pyramid have?' [WAIT TIME: 10 seconds. Cold-call 1 learner.] Build the full formula step-by-step. Then say: 'Now check your formula against the | Guided algebraic derivation anchored to the physical net (Face-by-Face Formula Building). By summing concrete measured areas before writing the generalised formula, learners see the formula as a compressed description of something they have already done physically — not an arbitrary rule to memorise. The A0-sheet extension problem connects the mathematics to real procurement decisions (how much material to order), directly mirroring the shopkeeper's claim in the phenomenon. | Observable indicators: (a) Learner exercise books show the step-by-step summation (Base area + 4 triangles) before the compact formula — not just a copied formula. (b) Numerical answer of 360 $cm^2$ for the square pyramid verification. (c) In the application problem, learners correctly use slant height (not vertical height) and correctly handle the 'open base' variation. (d) Exit check: teacher asks 2 learners at random to state in one sentence 'what slant height is and why we use it instead of the vertical height.' Correct response: 'Slant height is the height of the triangular face measured along the face from base edge to apex — it is what we physically see on the net, not the height going straight down inside the pyramid.' |

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| | Nairobi makes pyramid-shaped gift boxes with a square base of side 15 cm and a slant height of 20 cm. How many square centimetres of cardboard are needed for one box (open base — no base included)? If the shop makes 200 boxes, how many full A0 sheets of cardboard (84.1 cm × 118.9 cm) must they order, assuming 10% wastage?' | numbers you measured. Does 360 cm² match what you got by summing your table?' Allow 2 minutes for individual checking. Cold-call 2 learners to confirm or question. For the rectangular base, ask: 'Why can't we just write 4 × (½ × l × s) here? Discuss with your partner for 30 seconds.' [WAIT TIME: 30 seconds.] Cold-call 2 pairs. Guide the derivation of the two-pair formula. For the application problem, say: 'Solve this in your exercise book. You have 6 minutes. I will cold-call two learners to present their working on the board.' Circulate and note errors (common: forgetting to add the base, or using vertical height instead of slant height). After presentations, ask the class: 'If the gift box has no base because it sits on a table, which formula changes and how?' [WAIT TIME: 12 seconds. Cold-call 1 learner.] | | |
| **Phase 4 — Driving Question Board (DQB) Update**<br>📹 **VIDEO:** [Volume of a Pyramid - Overview]<br>Source: CK-12<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | Each learner writes one sticky note (or a designated section in their exercise book titled 'DQB Update') responding to TWO prompts: (1) 'What new evidence do I now have that helps answer: how much material is needed to make a pyramid-shaped object?' (2) 'Does today's evidence support or challenge the hardware-shop owner's claim that the frustum uses the most metal?' Groups share with the class and the teacher physically adds notes to the class DQB under the column 'Pyramid Evidence.' The class also updates a running comparison table on the DQB: Shape \| Formula \| TSA for our example dimensions \| Compared to cone (Lesson 1). | Direct learners to the class DQB (a section of the board or a large chart paper left up throughout the unit): 'In Lesson 1, we added cone evidence. Today we add pyramid evidence. I need two things from you on your sticky note: first, the formula you derived and a worked example number; second, whether today's lesson moves you closer to answering whether the frustum really does use the most metal.' Allow 3 minutes of individual writing. Then invite 4 learners (cold-call, different groups from Lesson 1 reporters) to read their note aloud and place it on the DQB. After all notes are placed, ask the whole class: 'Looking at our DQB so far — cones from Lesson 1, pyramids from today — what shape of question is | Driving Question Board as cumulative evidence tracker. The comparison table makes the unit's progression visible and tangible — learners can literally see the gap (frustum formula not yet known) that the later lessons will fill. This sustains curiosity and shows learners that each lesson is a deliberate piece of evidence, not an isolated topic. | Observable indicator: Each learner's DQB sticky note contains (a) a correct formula or numerical result and (b) a genuine connection to the phenomenon question (not just a restatement of the lesson topic). Teacher reviews 5–6 sticky notes as they are placed; if notes lack a connection to the phenomenon, pause and model one exemplar note aloud. |

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| | | emerging that we still cannot answer?' [WAIT TIME: 15 seconds.] Accept 2–3 responses. Prompt if needed: 'We know cone formula and pyramid formula — but do we yet know the frustum formula? Where does that appear in our sequence?' Point to 'frustum' lesson in the sequence visible on the DQB. Say: 'That question is now officially on our DQB as an open question.' | | |
| **Phase 5 — Model Building**<br>📹 **VIDEO:** [Volume of a Pyramid - Overview]<br>Source: CK-12<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | Learners revise their unit conceptual model — a diagram they began in Lesson 1 showing 'how shape affects the amount of material needed.' In Lesson 1, the model included only the cone with its net and formula. Now learners add: (a) a labelled sketch of the pyramid net showing base side l and slant height s; (b) the formula TSA = $l^2 + 2ls$ (square base) written next to the net; (c) an arrow and annotation comparing the pyramid's surface area to the cone's from Lesson 1 ('For our example, pyramid TSA = ___ cm² vs. cone TSA = ___ cm² — which is larger?'); (d) a 'Still unknown' box where learners write: 'We do not yet know the frustum formula — Lessons 5–6 will add this.' Learners complete the model revision individually in 5 minutes, then briefly share with their group. | Return learners' Lesson 1 model pages (or direct them to that page in their exercise books). Say: 'Your model is like a living document — a jua kali craftsman's reference sheet that gets updated every time they learn a new shape. Today you are adding the pyramid page.' Project a blank model template on the board and add the pyramid net sketch live as a demonstration: 'I am drawing the net, labelling l and s, and writing the formula — now you do the same in your book, but also fill in the actual numbers from today's worked example.' [WAIT TIME: Allow 5 minutes of silent individual work.] Circulate and check: (a) Is the slant height labelled on the triangular face (not the vertical height)? (b) Is the comparison to the cone present? After 5 minutes, cold-call 2 learners to describe their model revision to the class: 'Tell me one thing you added and one thing you still have as an open question.' Affirm the 'Still unknown' box: 'That open-question box is exactly what drives us to Lesson 3 and beyond. A good scientist and a good engineer always know what they do NOT yet know.' | Cumulative model revision (progressive model-building). The model serves as a cognitive anchor that grows lesson by lesson, helping learners see the unit as one coherent investigation rather than disconnected formula lessons. The explicit 'Still unknown' annotation maintains productive intellectual tension and directly links to the frustum mystery at the heart of the phenomenon. | Observable indicators for model quality: (a) Net sketch is present and labelled with correct dimension names (l = base side, s = slant height — not h). (b) Formula is written in algebraic form (not just numbers). (c) A numerical comparison to the cone from Lesson 1 is present (even if approximate). (d) 'Still unknown' box contains a specific open question about the frustum or sphere. Teacher collects 5 exercise books at random at the end of class to review model pages; findings inform the opening review in Lesson 3. |

| D. TEACHER REFLECTION |
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| After the lesson, reflect on the following questions:<br>1. SLANT HEIGHT MISCONCEPTION: How many learners pointed to the vertical height of the 3D pyramid instead of the slant height on the net during the traffic-light check? If more than 25% of the class showed confusion, plan a 5-minute warm-up at the start of Lesson 3 using a labelled diagram and a 'spot-the-error' question where the vertical height is incorrectly substituted into the formula.<br>2. NET CONSTRUCTION ACCURACY: Were learners able to construct the net accurately enough that it folded into a closed pyramid? If more than one group's net did not close, consider pre-cutting the nets for Lesson 3 (sphere/hemisphere) and reserving the construction activity for a longer double period.<br>3. FORMULA DERIVATION vs. MEMORISATION: Did learners sum face-by-face in their books before writing the compact formula, or did they copy the formula from the board directly? If the latter, the concrete-to-abstract bridge was not effective — restructure Explain phase to withhold the formula from the board until after learners have computed the numerical sum themselves.<br>4. DQB ENGAGEMENT: Were learners' sticky notes genuinely connecting to the phenomenon (frustum comparison) or were they just restating the formula? Low-quality DQB notes suggest learners are treating this as an isolated topic — re-read the phenomenon aloud at the start of Lesson 3 and explicitly ask 'What does today's lesson have to do with the Kirinyaga Road hardware shop?'<br>5. PACING: The 80-minute double lesson is tight. If net construction took more than 20 minutes, consider distributing pre-scored (but uncut) nets next time and using the time saved for the application problem extension (A0-sheet procurement calculation), which most strongly connects to the driving question. |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
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| **What did I observe?** | Learners cut out and folded cardboard nets of square-based and rectangular-based pyramids, labelled all five faces, and measured base dimensions and slant heights. They recorded measurements in a face-by-face area table and verified that the net folded into a closed 3D pyramid. |
| **What did I learn?** | The total surface area of a square-based pyramid is TSA = $l^2 + 2ls$ (base area plus four triangular faces). For a rectangular-based pyramid, opposite triangular faces form congruent pairs, giving TSA = $lb + ls_1 + bs_2$. Slant height (measured along the triangular face) must be used — not the vertical height of the pyramid. For a square pyramid with $l = 10$ cm and $s = 13$ cm, TSA = 360 $cm^2$. |
| **How does this explain the phenomenon?** | The formula is derived by summing the area of the square (or rectangular) base and the areas of all four triangular faces — the same process a jua kali sheet-metal worker in Kamukunji uses when pricing how much flat metal sheet to cut before hammering a pyramid-shaped object. This adds a second data point (after the cone in Lesson 1) to the class comparison of material use across different solid shapes, advancing the investigation of the hardware-shop owner's claim about the frustum. |

| LESSON 3 (80 minutes): Surface Area of a Sphere | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Learners derive and apply the formula SA = $4\pi r^2$ for the surface area of a sphere and hemisphere, connecting the concept to the anchoring phenomenon of how shape determines the amount of material needed to construct a solid. |
| **Knowledge** | The learner knows and understands that the surface area of a sphere is exactly four times the area of its great circle (SA = $4\pi r^2$), and that the surface area of a hemisphere is $3\pi r^2$ (curved surface $\pi r^2 \times 2$ plus the circular base $\pi r^2$). |
| **Skills** | The learner can: (a) estimate the diameter of spherical objects using string and ruler; (b) demonstrate experimentally that the surface area of a sphere equals four great-circle areas; (c) derive and apply SA = $4\pi r^2$ and SA_hemisphere = $3\pi r^2$ to solve real-life problems such as painting a mosque dome. |
| **Attitudes** | The learner appreciates the surprising elegance of the relationship between a sphere's surface area and its great circle, and recognises that understanding surface area has genuine economic and practical importance — as illustrated by the hardware shop phenomenon on Kirinyaga Road. |
| **Key Inquiry Question** | How are solids used in real life? |
| **Purpose in Storyline** | In Lesson 1 learners explored nets of cones and pyramids; in Lesson 2 they computed surface areas of cylinders and cones. Now in Lesson 3 they encounter the sphere — a shape with no flat net — and must discover a new experimental pathway to its surface area formula. This directly advances the driving question: the shopkeeper's three buckets all had flat or slanted faces; a spherical container would present an entirely different material-cost calculation, deepening learners' toolkit for comparing solids. |
| **Safety Notes** | When cutting oranges, learners must use care with knives/cutters. Teacher should pre-cut oranges into halves before the lesson if sharp implements are restricted. Remind learners to wash hands after handling fruit. String should be non-elastic for accurate measurement. |

| B. LESSON OVERVIEW |
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| Learners begin by handling spherical objects — soccer balls, oranges, and marbles — to estimate diameter using string and ruler, then predict the surface area using only their measurements. They then carry out the 'orange-peel flattening' activity: peeling an orange and pressing the peel pieces flat to cover four circles of the same radius as the orange's great circle, discovering experimentally that SA = $4\pi r^2$. After sense-making and formal derivation, learners apply the formula to calculate the amount of paint needed to coat a dome on a mosque in Mombasa, and the metal sheet needed for a spherical water tank in Kisumu. The lesson closes with model-building: learners add the sphere formula to their growing surface-area reference model and revisit whether a spherical bucket would be cheaper or more expensive than the frustum bucket in the anchoring phenomenon. |

| C. LESSON IMPLEMENTATION FRAMEWORK | | | | |
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| **Phase** | **Learner Experience** | **Teacher Moves** | **Sensemaking Strategy** | **Formative Assessment Strategy** |
| **Phase 1 — Predict** | Each group of 4 receives one orange, one marble, and a soccer ball (or printed photograph of one | Distribute materials before learners enter or during settling. Open with: 'Last lesson we | Strategy: Prediction Anchoring. By committing a numerical prediction to paper before instruction, | Observable indicator: Every learner has written a numeric prediction (even if incorrect) and a |

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| 📹 **VIDEO:** [Global winds and currents]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | with scale bar). Learners use a piece of string to measure the circumference of each object, calculate the radius, then write a personal prediction in their exercise books: 'I think the total surface area of my orange is ___ cm² because ___.' Learners also revisit the anchoring phenomenon: 'The shopkeeper has a cylindrical bucket, a frustum bucket, and a conical bucket — all holding 10 litres. If a fourth bucket were perfectly spherical, where do you think its surface area would rank — smallest, middle, or largest? Why?' Predictions are written silently for 3 minutes, then shared with a partner. | calculated how much sheet metal goes into a cone and a cylinder. Today we tackle a shape that cannot be unfolded into a flat net at all — the sphere. Before I tell you anything, I want you to measure and predict.' Circulate silently for 3 minutes while learners write predictions — do NOT answer questions about formulas. After 3 minutes, cold-call 3 learners: 'Achieng, what circumference did you measure for your orange?' [WAIT 10 seconds] 'Kamau, how did you convert that to a radius?' [WAIT 12 seconds] 'Fatuma, what surface area did you predict, and what was your reasoning?' [WAIT 15 seconds]. Record three different predictions on the board without evaluating them. Say: 'We will return to these predictions at the end of Phase 2 to see who was closest — and more importantly, why.' | learners activate prior knowledge of circles and circumference (from earlier sub-strands) and create a cognitive 'gap' that the experiment will close. Writing the prediction before peer discussion ensures individual accountability. | written justification involving radius or circumference. Teacher scans books during circulation — if more than one-third of learners leave the justification blank, pause and prompt: 'Use $C = 2\pi r$ to find r first, then think about what you already know about circles.' |
| **Phase 2 — Observe**<br>📹 **VIDEO:** [Global winds and currents]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | ACTIVITY — The Orange-Peel Great-Circle Experiment. Each group receives: one orange (pre-halved by teacher if necessary), string, ruler, four small sheets of plain paper, a marker pen, and a plate. Step 1 — Trace the great circle: Learners place the flat cut face of the orange half on paper and trace the circular outline. This is the 'great circle.' They measure its radius and compute its area ($A = \pi r^2$), recording in their books. Step 2 — Peel and flatten: Learners carefully peel the orange skin in small pieces and press them flat onto copies of the great circle drawn on paper, filling the circles with peel. They observe how many great-circle drawings are completely covered. Step 3 — Record evidence: Learners record: | Introduce the activity: 'You are going to do what mathematicians call an empirical investigation — you will use physical evidence to discover a formula. Your job is to observe carefully and record exactly what you see, not what you think should happen.' Circulate during peeling. When a group covers approximately 4 circles, ask: 'How many circles did your peel cover?' [WAIT 10 seconds]. Cold-call 4 different groups for their count — write each on the board (expected answers: approximately 4). Ask: 'Nakuru group got 3.8, Mombasa group got 4.1 — why are these not exactly 4?' [WAIT 15 seconds — accept: peel stretches, measurement error, orange is not a perfect sphere]. After all groups report, | Strategy: Evidence-to-Formula Bridging via Concrete Manipulation. The orange-peel activity gives learners a tactile, observable experience of the abstract formula. Recording the number of great circles covered creates a data point that the class aggregates, mimicking scientific practice (NGSS Science & Engineering Practice 3: Planning and Carrying Out Investigations; Practice 4: Analysing and Interpreting Data). The variation across groups (3.8–4.2) is deliberately used to discuss measurement error and the distinction between empirical result and mathematical proof. | Observable indicator: Each group's exercise book shows (a) a traced great circle with radius labelled, (b) a numeric count of circles covered (should be 3.8–4.2), and (c) a revised surface area estimate using $4\pi r^2$. Teacher checks 6 books (one per group). Red flag: if any group recorded fewer than 3 or more than 5 circles, check their orange was not unusually shaped and their circle tracing was from the true great circle (equatorial cross-section). |

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| | 'We covered ___ great circles with the peel from one orange.' Step 4 — Digital extension (if devices available): Learners watch a 90-second GeoGebra animation showing a sphere 'unrolling' onto four circles, accessible via Kenya Education Cloud. Step 5 — Learners return to their numeric predictions from Phase 1 and annotate: 'My prediction was ___ $cm^2$. The experiment suggests SA ≈ 4 × $\pi r^2$ = ___ $cm^2$.' | say: 'Mathematicians have proven — not just observed — that the answer is exactly 4. So SA = 4 × area of great circle = $4\pi r^2$.' Write this prominently on the board. Then ask: 'Achieng, how does this compare to your prediction from Phase 1?' [WAIT 10 seconds]. Do not move on until at least 5 learners have verbally connected the experiment result to their earlier prediction. | | |
| **Phase 3 — Explain**<br>📹 **VIDEO:** Global winds and currents<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | PART A — Formula Derivation and Notation. Learners copy and complete a structured derivation frame in their books: 'Surface area of a sphere = ___ × area of great circle = ___ × $\pi r^2$ = ___.' They then derive the hemisphere formula: curved surface area = ½ × $4\pi r^2$ = $2\pi r^2$; total surface area of closed hemisphere = $2\pi r^2 + \pi r^2 = 3\pi r^2$. PART B — Worked Examples (Kenya Context). Example 1 (Mosque Dome, Mombasa): The Mandhry Mosque near the Old Port in Mombasa has a hemispherical dome with diameter 8.4 m. A painter charges Ksh 350 per $m^2$ to paint the curved outer surface. Calculate: (i) the curved surface area of the dome; (ii) the total cost of painting it. Learners work in pairs for 5 minutes, then one pair presents on the board. Example 2 (Spherical Water Tank, Kisumu): A spherical water tank in a Kisumu estate has a radius of 1.5 m. Calculate the total surface area of sheet metal needed to construct it. How does this compare to the cylindrical bucket in our phenomenon (which has the same 10-litre capacity)? Learners solve individually (5 minutes). PART C — Connecting | For Part A: 'Before I show you the formal derivation, I want YOU to write it using only what we observed — fill in the blanks.' [Allow 2 minutes silent work]. Cold-call 2 learners to read their completed derivation aloud [WAIT 10 seconds each]. For the hemisphere: 'A hemisphere is half a sphere — but is its surface area exactly half of $4\pi r^2$? Think carefully — what surface does a cut hemisphere have that a whole sphere does not?' [WAIT 15 seconds]. Cold-call 3 learners [expect: the flat circular base]. Write both formulas on the board in a coloured box. For Example 1: 'Work with your partner — you have 5 minutes. I will cold-call one pair to present every step.' Circulate and identify a pair that shows unit conversion (diameter to radius) correctly. For Example 2: 'This one is individual — 5 minutes, silent.' After marking, ask: 'Omondi, which bucket — cylinder or sphere — needs more sheet metal for 10 litres? What does that tell the shopkeeper?' [WAIT 15 seconds]. Ensure the class hears that the sphere, for a given volume, has the minimum possible surface area — a key insight for | Strategy: Structured Derivation Frames + Real-World Problem Solving. The fill-in-the-blank derivation frame (NGSS Practice 6: Constructing Explanations) requires learners to articulate the logical chain rather than copy a completed proof. The Mombasa and Kisumu contexts ground the formula in Kenyan economic reality (cost of painting, cost of sheet metal), making the mathematics purposeful. Connecting the sphere to the anchoring phenomenon creates conceptual continuity across the 8-lesson sequence. | Observable indicator for Part A: Every learner's book shows SA = $4\pi r^2$ and $SA\_hemisphere = 3\pi r^2$ written in their own derivation frame (not just copied from board). Observable indicator for Examples: Learners correctly substitute r (not d) into the formula — the most common error is using diameter directly. Teacher checks 8 books during individual work on Example 2. If more than 3 learners use d instead of r, stop the class: 'Check your substitution — what does r stand for in our formula?' |

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| | to the Phenomenon. Learners answer in their books: 'If the shopkeeper on Kirinyaga Road added a fourth bucket — a spherical one holding exactly 10 litres — how much sheet metal would it need? Is the shopkeeper correct that the frustum uses the most metal, or does the sphere challenge that claim?' | the driving question. | | |
| **Phase 4 — Driving Question Board (DQB) Update**<br>📹 **VIDEO:**<br>Global winds and currents<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Each learner receives a sticky note (or tears a small piece of paper). They write ONE updated understanding and ONE remaining question in the following format: 'I now understand: ___' and 'I am still wondering: ___'. Examples of expected responses — Understanding: 'A sphere's surface area is 4 times its great circle, so SA = $4\pi r^2$'; Wondering: 'Why does the sphere have the smallest surface area for a given volume?' or 'How do you find the surface area of a frustum that is almost spherical?' Learners post their notes on the class Driving Question Board under two columns: WHAT WE NOW KNOW and WHAT WE STILL WONDER. The teacher reads 4–5 notes aloud and clusters them. The class votes (hands up) on the top 2 'still wondering' questions to carry into Lessons 4–5. | Say: 'The shopkeeper on Kirinyaga Road said the frustum uses the most metal. After today's lesson, can we confirm or challenge that? Write what you now know and what you still need to figure out.' [Allow 3 minutes silent writing]. As learners post notes, read aloud: 'Wanjiku writes: I now know SA = $4\pi r^2$. She is still wondering whether a sphere could replace the frustum and cost less. What do others think?' [WAIT 12 seconds — cold-call 2 learners]. After clustering, say: 'These two questions — about frustum surface area and about comparing shapes — are exactly what Lessons 4 and 5 will answer. You are doing what real engineers and shopkeepers do: build understanding piece by piece.' Photograph the DQB (or have a learner sketch it) for continuity. | Strategy: Driving Question Board Metacognitive Update (NGSS Practice 8: Obtaining, Evaluating and Communicating Information). The DQB makes learning visible and cumulative. By separating 'now know' from 'still wonder', learners practise metacognition — monitoring their own understanding. The teacher's act of reading notes aloud and clustering validates all contributions and models scientific discourse. | Observable indicator: Every learner produces a written note with both a knowledge statement and a question. Red flag: if learners write only vague statements ('I learnt about spheres'), prompt: 'Write the actual formula you can now use, and a specific number you could calculate.' Teacher photographs or collects notes to review before Lesson 4. |
| **Phase 5 — Model Building**<br>📹 **VIDEO:**<br>Global winds and currents<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos | Learners retrieve their cumulative Surface Area Reference Model — a hand-drawn or template-based table begun in Lesson 1 that records: Solid \| Formula \| Net/Diagram \| Kenya Example. They add the sphere and hemisphere rows, completing all columns. Then, as a group challenge, they calculate surface areas for all four 'buckets' in the | Distribute or direct learners to open their cumulative model tables. Say: 'Add the sphere and hemisphere rows now — use the formulas we derived, draw the shape, and write a Kenya example.' [Allow 4 minutes]. Circulate and check that diagrams show radius labelled correctly and that hemisphere diagram shows both curved surface and flat base. | Strategy: Cumulative Model Revision (NGSS Practice 2: Developing and Using Models). The reference model table functions as a living scientific model that grows with each lesson. Adding the sphere row while leaving the frustum row deliberately blank creates anticipatory tension — learners know their comparison is | Observable indicator: Learner model tables contain correct formulas for sphere ($4\pi r^2$) and hemisphere ($3\pi r^2$), a labelled diagram, and a Kenyan example (e.g., mosque dome, water tank). The comparison table shows a numeric surface area value for the sphere holding 10 litres (expected: SA ≈ 2,257 cm²). Teacher collects 6 model tables (one per group) at |

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| 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings | phenomenon: Cylinder (from Lesson 2), Cone (from Lesson 2), Sphere (today), and leave the Frustum row blank with the label 'Coming in Lesson 4.' Groups compare their four values and write a ranked list: 'From least to most sheet metal: ___.' Finally, each learner writes a one-sentence 'model statement': 'For a fixed volume of 10 litres, the [shape] requires the least sheet metal because ___.' This statement will be revised as more evidence is gathered. | After 4 minutes: 'Now, in your groups, use the 10-litre volume to find the radius of a sphere holding 10 litres.' [$V = (4/3)\pi r^3 \rightarrow r \approx 13.4$ cm]. Write the setup on the board — do NOT solve it for them. Cold-call one group: 'Kisumu group — what radius did you get?' [WAIT 15 seconds]. Then: 'Now compute SA $= 4\pi r^2$ for that radius and add it to your comparison table.' Close by asking: 'Based on what we know so far — three of the four shapes — is the shopkeeper's claim about the frustum looking plausible or unlikely? Turn and tell your partner in one sentence.' [WAIT 20 seconds for partner talk]. Cold-call 3 pairs. Record their sentences on the board — these become the class's current working hypothesis to test in Lesson 4. | incomplete, motivating engagement with Lesson 4. The ranked-list activity requires learners to synthesise evidence across three lessons, building the cross-lesson coherence that is central to the phenomenon-driven storyline. | end of lesson to check for accuracy and use as entry-point data for Lesson 4. |

## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the orange-peel activity produce counts close enough to 4 (between 3.7 and 4.3) in most groups? If not, was this due to orange shape, elastic peel, or imprecise great-circle tracing — and how will you address measurement technique in Lesson 4? (2) Were learners able to independently derive the hemisphere formula ($3\pi r^2$) by reasoning about the flat base, or did most require prompting? If prompting was needed, build an explicit 'decomposing a solid into faces' scaffold into Lesson 4 for the frustum. (3) Did learners successfully calculate the radius of a sphere with V = 10 litres using cube-root reasoning, or did the unfamiliar rearrangement stall groups? Note which groups struggled — they may need a bridging question at the start of Lesson 4. (4) Review the DQB sticky notes: how many learners articulated a specific formula in their 'now know' statement versus a vague paraphrase? This ratio indicates the depth of conceptual consolidation achieved. (5) Is the class's working hypothesis about the shopkeeper's claim (frustum vs. sphere) becoming more evidence-based, or are learners still guessing? Use the model table entries to plan targeted cold-call questions at the start of Lesson 4.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| **What did I observe?** | Learners peeled an orange and pressed the peel flat to cover circles traced from the orange's great circle (equatorial cross-section), observing that the peel filled approximately 4 such circles. |
| **What did I learn?** | The surface area of a sphere equals exactly 4 times the area of its great circle: SA $= 4\pi r^2$. The total surface area of a closed hemisphere is $3\pi r^2$ (curved surface $2\pi r^2$ plus circular base $\pi r^2$). |
| **How does this explain the phenomenon?** | Learners applied SA $= 4\pi r^2$ to calculate the curved surface area and painting cost of a hemispherical mosque dome in Mombasa, and computed the sheet metal needed for a spherical water tank in Kisumu. They compared the sphere's surface area to the cylinder and cone from Lessons 1–2 for the same 10-litre volume, advancing the driving question about which bucket shape uses |

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| | the most material. |

| LESSON 4 (80 minutes): Surface Area of a Hemisphere | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Learners extend their understanding of curved surface area from the full sphere (Lesson 3) to the hemisphere, deriving and applying the formulas for curved surface area ($2\pi r^2$) and total surface area ($3\pi r^2$) in real-life Kenyan contexts such as hemispherical safari tents at the Maasai Mara. |
| **Knowledge** | By the end of the lesson, the learner should be able to: (a) derive the curved surface area of a hemisphere as $2\pi r^2$, (b) calculate the total surface area of a hemisphere as $3\pi r^2$, (c) apply both formulas to real-life situations such as estimating material needed for hemispherical structures. |
| **Skills** | The learner is able to: (a) slice a spherical model to identify the hemisphere and its two components (curved surface and flat circular face); (b) use nets and measurements to verify surface area formulas; (c) solve multi-step surface area problems connecting hemispheres to the anchoring phenomenon. |
| **Attitudes** | The learner appreciates how the shape of a solid — even when volume is held constant — fundamentally changes the amount of material needed to construct or cover it, reinforcing curiosity about the three buckets in the anchoring phenomenon. |
| **Key Inquiry Question** | How are solids used in real life? |
| **Purpose in Storyline** | Building on Lesson 3 (full sphere surface area = $4\pi r^2$), learners slice a spherical model in half and discover that the hemisphere has TWO surface components: a curved surface (half of $4\pi r^2 = 2\pi r^2$) and a flat circular face ($\pi r^2$), giving a total of $3\pi r^2$. Learners apply this to estimating canvas for a hemispherical safari tent at Maasai Mara, then return to the DQB at mid-unit to update their predictions about the three buckets — connecting hemisphere reasoning to the frustum's unexpectedly large surface area. |
| **Safety Notes** | When cutting or handling physical spherical models (e.g., halved oranges or rubber balls), learners should be careful with cutting tools. Wash hands after handling fruit models. Ensure shared materials such as rulers and compasses are handled and passed safely between peers. |

## B. LESSON OVERVIEW

In this fourth lesson of the evidence-gathering sequence, learners build directly on their Lesson 3 knowledge of the full sphere (SA = $4\pi r^2$) by physically or visually slicing a spherical object in half to create a hemisphere. Through guided observation of a halved orange or rubber ball, learners identify the two distinct surface components of a hemisphere: the curved outer surface and the flat circular base. They derive the curved surface area formula ($2\pi r^2$) as exactly half of the full sphere, then add the area of the circular face ($\pi r^2$) to obtain the total surface area formula ($3\pi r^2$). The lesson anchors these formulas in a vivid Kenyan context — estimating the canvas needed to cover a hemispherical tent at a Maasai Mara safari camp — before returning learners to the Driving Question Board (DQB) for a critical mid-unit reflection: how does this hemisphere reasoning help explain why the frustum-shaped bucket uses more metal than the cylinder? This is the mid-unit formative checkpoint for the 8-lesson sequence.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **1. Predict Phase**<br>📹 **VIDEO:**<br>Global winds and currents<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Learners are shown a large orange (or rubber ball) and the teacher then holds up a knife (or pre-cut halved model) and asks: 'If I slice this sphere exactly in half, what surfaces will I see on the cut piece? How many surfaces does a hemisphere have — and how would you estimate the total area of material needed to wrap the outside of this half-sphere completely?' Learners individually sketch the hemisphere on paper, label the surfaces they predict exist, and write a numerical estimate for the total surface area if the radius is 7 cm. They record their predictions in their exercise books before any instruction is given. | Display the whole orange or spherical model prominently. Ask: 'Look carefully — if I cut this exactly in half, what surfaces will the cut piece have?' Allow 12 seconds of WAIT TIME. Cold-call 3 learners to share their predictions. Ask a follow-up: 'Amina says there is only one curved surface. Chebet says there are two surfaces — a curved one and a flat one. Who do you agree with, and why?' Allow 10 seconds WAIT TIME then cold-call 2 more learners. Instruct: 'Open your exercise books. Sketch a hemisphere, label all the surfaces you predict exist, and estimate the total surface area if radius = 7 cm. You have 4 minutes — work silently and independently.' | Predict–Observe–Explain (POE) Activation: By committing a prediction to paper before instruction, learners activate their prior knowledge of sphere surface area from Lesson 3 and create a personal cognitive anchor. When the observation phase later confirms or challenges their prediction, the contrast drives deeper encoding of the hemisphere's two-component structure. | Teacher circulates and scans sketches. Observable indicators of prior understanding: (1) learner correctly draws both a curved outer surface AND a flat circular base — indicates readiness; (2) learner draws only a curved surface — indicates the flat face concept needs explicit attention in the Observe phase; (3) learner's estimate is in the range 200–350 $cm^2$ for $r = 7$ cm — indicates number-sense alignment with the correct answer of $3\pi(7^2) \approx 462$ $cm^2$. Note which learners need targeted support. |
| **2. Observe Phase**<br>📹 **VIDEO:**<br>Global winds and currents<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | In pairs, learners receive a halved orange (or pre-cut rubber ball model) and a printed net diagram of a hemisphere. They complete three structured observations: (A) They physically trace the flat circular face of the halved orange onto paper and measure its radius, then calculate its area ($\pi r^2$). (B) They carefully peel the curved orange skin and attempt to flatten it, observing that it does not flatten perfectly — linking this to the earlier sphere peel activity in Lesson 3 and to the formula $2\pi r^2$ for the curved surface. (C) They examine the printed hemisphere net (curved surface unrolled as a semicircle of appropriate dimensions alongside the circular base) and annotate which part corresponds to which formula. Learners record all measurements, sketches and observations in a structured table in their exercise books. | Distribute halved oranges/models and printed net diagrams. Say: 'You have exactly 15 minutes for three observations — I will call time at each stage.' Stage A (5 min): 'Trace the flat face onto your paper. Measure the radius. Calculate its area using $\pi r^2$. Record in your table.' Circulate; prompt with: 'What shape is that flat face? What is its area formula?' Stage B (5 min): 'Now carefully peel the curved skin. Try to lay it flat. What do you notice?' Allow 10 seconds WAIT TIME after the action, then cold-call 3 pairs: 'What happened when you tried to flatten it — and what does that remind you of from Lesson 3?' Stage C (5 min): 'Look at your printed net. Point to the part that represents the curved surface. Point to the part that represents the flat face. Annotate both parts with the correct area formula.' Cold-call 2 pairs to share | Concrete-Representational-Abstract (CRA) Progression: Learners begin with a concrete physical object (halved orange), move to a semi-representational net diagram (printed template), and are guided toward the abstract formula. This three-stage progression ensures the formula $2\pi r^2$ is understood as arising from the physical object, not memorised in isolation. The peeling activity directly parallels the full-sphere peeling from Lesson 3, building a powerful conceptual bridge. | Check each pair's structured table for: (1) Correct identification of the flat face as a circle with area = $\pi r^2$ — a non-negotiable prerequisite for the Explain phase; (2) Recorded observation that the curved skin does not flatten perfectly, confirming they connect the curved surface to the sphere formula; (3) Correct annotation of both net components. Pairs who annotate only one component or misidentify the flat face shape should be flagged for targeted questioning during the Explain phase. |

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| | | annotations. Pose the bridging question: 'In Lesson 3 we found the full sphere has surface area $4\pi r^2$. If this hemisphere is exactly half a sphere, what do you predict the curved surface area formula is?' | | |
| **3. Explain Phase**<br>📹 **VIDEO:** Global winds and currents<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | The teacher leads a whole-class derivation, with learners contributing each step. Learners first confirm that the curved surface of a hemisphere = ½ × (surface area of full sphere) = ½ × $4\pi r^2$ = $2\pi r^2$. They then add the flat circular base area $\pi r^2$ to obtain the total surface area = $2\pi r^2 + \pi r^2 = 3\pi r^2$. The class then works through two graded problems: Problem 1 — A hemispherical bowl used to serve ugali at a Nairobi restaurant has a radius of 14 cm. Calculate: (a) the curved surface area, (b) the total outer surface area. Problem 2 — A hemispherical canvas tent at a Maasai Mara safari camp has an internal radius of 3.5 metres. The camp manager wants to know how many square metres of canvas are needed to make the tent (curved surface only, since the tent sits on the ground). Calculate the canvas required and estimate the cost if canvas costs Ksh 850 per square metre. Learners work individually on both problems, then compare answers with a partner before a whole-class discussion. | Open with: 'From your observations, Lemayian says the curved surface area of a hemisphere is half of $4\pi r^2$. Do you agree? What does that simplify to?' Allow 10 seconds WAIT TIME. Cold-call 3 learners. Write the derivation step-by-step on the board: 'Curved SA = ½ × $4\pi r^2$ = $2\pi r^2$. Now — the hemisphere also has a flat face. What shape is it?' Cold-call 2 learners. 'What is its area?' Cold-call 2 learners. Write: 'Total SA = $2\pi r^2 + \pi r^2 = 3\pi r^2$.' Pause and ask: 'Why is the total $3\pi r^2$ and not $2\pi r^2$? What is the extra $\pi r^2$ representing physically?' Allow 12 seconds WAIT TIME. Cold-call 3 learners. Transition to Problem 1: 'Work Problem 1 individually — you have 5 minutes.' Circulate, check working. After 5 min, cold-call one learner to present their working for part (a) and another for part (b). Correct misconceptions immediately. Transition to Problem 2: 'The safari tent — read carefully. Do we need the curved surface only, or total surface area? Why?' Allow 10 seconds WAIT TIME. Cold-call 2 learners before they begin calculating. After 7 minutes of individual work, instruct: 'Compare your answer with your partner. If you disagree, identify exactly which step differs.' Cold-call one pair to present. Close by asking: 'How does the hemisphere tent problem connect back to our three | Structured Derivation with Anchored Application: The teacher does not simply state the formula but facilitates a step-by-step derivation that learners can reconstruct independently. Each formula component is grounded in a physical referent (the orange skin = curved surface; the traced circle = flat face). The two application problems escalate in complexity and real-world embeddedness — the safari tent problem explicitly quantifies material cost in Kenyan Shillings, making the formula consequential rather than abstract. | During individual problem work, circulate and check: (1) Does the learner correctly use $2\pi r^2$ for the curved surface only and $3\pi r^2$ for total? — confusion between the two is the primary misconception; (2) Does the learner correctly substitute radius (not diameter) when given diameter information? — a common error; (3) In Problem 2, does the learner correctly identify that only the curved surface area is needed for a ground-based tent? — tests reading comprehension and contextual reasoning. At cold-call stage, listen for whether learners can articulate WHY $3\pi r^2$ rather than just calculating it. |

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| | | buckets on Kirinyaga Road? Is a hemisphere more or less material-efficient than a cylinder for a given volume?' | | |
| **4. Driving Question Board (DQB) Update — Mid-Unit Formative Check**<br>📹 **VIDEO:** Global winds and currents<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | This is the designated mid-unit DQB checkpoint (Lesson 4 of 8). Learners individually return to their original DQB sticky-note predictions about the three Kirinyaga Road buckets (cylinder, frustum, cone — all holding 10 litres). Each learner receives a new colour sticky note and responds to two prompts: (A) 'Using what you now know about surface areas of cones, pyramids, spheres and hemispheres, revise your prediction: which bucket do you now think uses the most metal — and why?' (B) 'What new questions has today's lesson raised for you about the frustum-shaped bucket?' Learners post their revised predictions and new questions on the DQB. The class briefly reviews the board together, and the teacher categorises emerging questions that will be addressed in Lessons 5–7. | Distribute new-colour sticky notes (different from Lesson 1 colour to show progression). Say: 'We are halfway through our investigation of the three buckets. You now know how to calculate the surface area of cones, pyramids, spheres and hemispheres. It's time to update your thinking.' Read Prompt A aloud: 'Write on your sticky note: which bucket do you now think uses the most metal — cylinder, frustum or cone — and give one mathematical reason.' Allow 5 minutes of silent individual writing. Read Prompt B aloud: 'Write one new question you now have about the frustum bucket — something you are still curious about.' Allow 3 minutes. Instruct learners to post both notes on the DQB. Scan the board and narrate patterns aloud: 'I notice many of you are now saying the frustum uses the most metal — but your reasons differ. Some say it is because the frustum has two different circular faces. Others say it has a slanted surface that is longer than a cone's slant. We will put both of these ideas to the test in Lesson 5.' Close with: 'What one word describes how you feel about the frustum right now — curious, confused, or confident?' Cold-call 5 learners by name. | Visible Thinking Routine — Claim–Support–Question (CSQ): Learners are asked to make a revised Claim (which bucket uses most metal), provide mathematical Support (one reason grounded in formulas learned so far), and pose a new Question (what they still need to investigate). This routine makes thinking visible to both teacher and learner, documents conceptual growth from Lesson 1 to Lesson 4, and generates the inquiry agenda for the remaining lessons. The two-colour sticky note system provides a concrete visual record of how understanding has evolved. | Read sticky notes immediately after posting. Key indicators: (1) Learners who now cite a specific formula (e.g., 'the frustum has two circles plus a curved surface') show formula-to-context transfer — these learners are ready for Lesson 5 frustum derivation; (2) Learners who still cannot differentiate between volume and surface area in their reasoning (e.g., 'the frustum holds the same water so it must use the same metal') need targeted intervention — note their names for small-group follow-up; (3) The quality and specificity of new questions on Prompt B reveals depth of curiosity and readiness for the frustum formula in Lesson 5. |
| **5. Model Building Phase**<br>📹 **VIDEO:** Global winds and currents | Learners return to their cumulative 'Surface Area Model' — a running annotated diagram they have been building since Lesson 1, showing all solids studied so far with their surface area formulas. Today, | Say: 'Open your cumulative Surface Area Model from your exercise book. Add the hemisphere today — draw it, label both surfaces, write both formulas. You have 6 minutes.' Circulate and | Cumulative Model Revision (Running Summary Diagram): The ongoing annotated diagram functions as a visible, student-generated knowledge structure that accumulates across all 8 | Scan cumulative models for: (1) Correct labelling of BOTH hemisphere surfaces — the most common omission is the flat circular face; (2) Correct formulas written adjacent to each labelled |

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| Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | learners add the hemisphere to the model: they draw a labelled hemisphere, annotate the curved surface ($2\pi r^2$) and the flat circular base ($\pi r^2$), and write the total surface area formula ($3\pi r^2$). They then add a written comparison statement connecting the hemisphere to the three buckets: 'A hemispherical bucket of radius r would have total SA = $3\pi r^2$. This is [more/less/the same] material as the cylindrical bucket we studied in Lesson 2, because...' Learners who finish early calculate the surface area of a hemisphere with the same volume as the 10-litre buckets (radius ≈ 13.4 cm) and compare it to the cylinder from Lesson 2. | check that learners distinguish $2\pi r^2$ (curved only) from $3\pi r^2$ (total) in their annotations. After 4 minutes, prompt: 'Now add the comparison statement. Does a hemisphere use more or less metal than the cylinder from Lesson 2? Write a sentence with a mathematical reason.' After 6 minutes, cold-call 2 learners to read their comparison statements. Address any remaining confusion between the curved surface area and total surface area formulas. Close the lesson: 'Our model is growing. In Lesson 5, we tackle the most complex shape yet — the frustum. Your DQB questions about why the frustum uses so much metal will finally be answered.' Assign homework: 'A hemispherical water tank in Kisumu has a diameter of 2.8 m. Calculate: (a) the curved surface area, (b) the total surface area including the flat base. Show all working.' | lessons. Adding the hemisphere forces learners to integrate the new formula into an existing framework (cone, pyramid, sphere), notice structural relationships (hemisphere total SA = ¾ of sphere SA), and articulate a comparison to the bucket phenomenon. This strategy promotes retention, revision readiness and cross-lesson coherence. | surface — not just the total formula; (3) A comparison statement that references a specific formula from a previous lesson — indicating cross-lesson knowledge integration. Collect 5–8 exercise books at random to review homework at the start of Lesson 5 as a spaced retrieval check. |

## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners consistently distinguish between the curved surface area formula ($2\pi r^2$) and the total surface area formula ($3\pi r^2$) — and could they explain in their own words what the extra $\pi r^2$ physically represents? If many learners conflated the two, consider opening Lesson 5 with a 3-minute retrieval prompt using a new radius value. (2) During the DQB mid-unit check, did the revised sticky-note predictions show genuine conceptual growth from Lesson 1, or were learners still anchoring reasoning in volume rather than surface area? If so, the phenomenon connection between surface area and material cost may need to be re-stated explicitly at the start of Lesson 5. (3) Was the Maasai Mara safari tent context sufficiently engaging to motivate the formula application, or did learners disengage during the cost-calculation step? Consider whether a more locally immediate context (e.g., a domed maize storage silo in the Rift Valley) might resonate better with your specific class. (4) Were there learners who were ready to extend to the hemisphere-frustum comparison during the Model Building phase? If so, design a stretch task for Lesson 5 that asks them to predict the frustum formula using hemisphere and cone reasoning before the class derivation.

## E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

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| **What did I observe?** | Learners sliced a spherical model (halved orange or rubber ball) in half and observed that a hemisphere has two distinct surface components: a curved outer surface and a flat circular base. They physically traced the flat face, measured its radius, peeled the curved surface, and examined a printed hemisphere net — confirming the two-component structure directly from a concrete object. |

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| **What did I learn?** | The curved surface area of a hemisphere = $2\pi r^2$ (exactly half the full sphere surface area of $4\pi r^2$). The total surface area of a hemisphere = $2\pi r^2 + \pi r^2 = 3\pi r^2$, where the additional $\pi r^2$ accounts for the flat circular base. These formulas were applied to calculate the canvas needed for a hemispherical safari tent at Maasai Mara and the cost in Kenyan Shillings. |
| **How does this explain the phenomenon?** | At the mid-unit DQB checkpoint, learners updated their predictions about the three Kirinyaga Road buckets using the surface area formulas learned so far (cone, pyramid, sphere, hemisphere). They articulated revised claims about which bucket uses the most metal and generated new specific questions about the frustum-shaped bucket's surface area — setting the inquiry agenda for Lessons 5 and 6. |

| LESSON 5 (80 minutes): Introduction to Frustums: Nets and Vocabulary | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Learners physically cut a conical model along a plane parallel to the base to produce a frustum, observe the two circular ends and slant height, unfold the frustum net, and derive and apply the curved surface area formula $\pi(R + r)l$ as well as the Pythagorean slant-height relationship $l = \sqrt{h^2 + (R - r)^2}$. |
| **Knowledge** | By the end of the lesson the learner should be able to: (a) identify the key dimensions of a frustum of a cone — large radius R, small radius r, vertical height h, and slant height l; (b) state the formula for the curved surface area of a frustum of a cone as $\pi(R + r)l$; (c) derive the slant height of a frustum using the relationship $l = \sqrt{h^2 + (R - r)^2}$; (d) recognise the frustum net as the difference between two sectors. |
| **Skills** | Cutting and constructing physical models; measuring dimensions of three-dimensional solids; unfolding and tracing nets; applying the Pythagorean theorem in a solid-geometry context; computing curved surface area of a frustum using the formula $\pi(R + r)l$. |
| **Attitudes** | Appreciate the utility of frustum geometry in everyday Kenyan objects (buckets, lampshades, grain silos); demonstrate curiosity and perseverance when the net of a frustum turns out to be larger and more complex than expected; show responsibility in handling cutting tools and sharing resources. |
| **Key Inquiry Question** | How are solids used in real life? |
| **Purpose in Storyline** | In Lessons 1–4 learners computed surface areas of cones, pyramids, spheres and hemispheres. Now they turn to the frustum — the tapered bucket the Kirinyaga Road shopkeeper claims uses the most metal. This lesson gives learners the conceptual vocabulary and the curved-surface-area formula they need to eventually verify or refute that claim. By cutting a cone and unfolding its frustum net they directly see why the formula involves two radii and a slant height, building the bridge to the full surface-area calculation in Lesson 6. |
| **Safety Notes** | Learners must handle craft knives, scissors and compasses under direct teacher supervision. Cutting must be done on a cardboard mat. No learner should pass a sharp instrument blade-first. Compass points should be capped when not in use. |

## B. LESSON OVERVIEW

Learners begin by recalling the conical bucket from the anchoring phenomenon and predicting what shape results when a cone is sliced parallel to its base. Each group then physically cuts a pre-scored paper cone along a marked parallel plane to reveal a frustum, measures its dimensions (R, r, h, l), and unfolds the net onto squared paper. Through guided questioning learners see that the curved surface of a frustum is the difference between two full cone-lateral-surface sectors, leading to the formula $\pi(R + r)l$. They also derive slant height l from the Pythagorean relationship $l = \sqrt{h^2 + (R - r)^2}$ and practise on two real-world Kenyan contexts: a tapered metal bucket sold on Kirinyaga Road (R = 18 cm, r = 12 cm, h = 30 cm) and a conical grain-storage silo frustum from Eldoret (R = 1.5 m, r = 0.6 m, h = 2.4 m). The lesson closes with a DQB update and a cumulative model revision connecting all solid shapes studied so far.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Phase 1 —** | Learners sit in groups of four. Each group has a paper cone (pre- | Display the three Kirinyaga Road buckets (physical models or clear | Prediction with accountability — learners commit a written | Observable indicator: every learner has at least two of the |

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| **Predict**<br>📹 **VIDEO:**<br>Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Volume of the Cone (PLIX)<br>Source: CK-12<br>🔍 Search ARES for similar readings | made, 15 cm tall, base radius 8 cm) with a faint pencil line drawn parallel to the base at 5 cm from the apex. Without cutting, learners individually sketch in their exercise books: (a) what the solid will look like after cutting along that line; (b) what they think the net (unfolded surface) of the leftover bottom piece will look like; (c) whether they think the leftover piece will need more or less paper to cover its curved surface than the original cone. Learners then share predictions with their group partner and record one agreed group prediction on a sticky note posted on the classroom 'Prediction Wall'. | photos on the board). Point to the frustum bucket and say: 'In Lesson 4 we found the curved surface area of the full cone. Today we slice that cone. Before we touch the scissors, I want your honest prediction.' Distribute the predict prompt slip (three sentence-starters printed on a half-page): 'The shape I will get is…', 'The net will look like…', 'The curved surface will be (more / less / the same) because…'. Allow 4 minutes of silent individual thinking — enforce NO talking during this period. After 4 minutes say: 'Turn to your partner — 60 seconds — compare predictions.' Cold-call 3 groups to read their sticky notes aloud. Do NOT correct any prediction; instead annotate them with a question mark and say: 'Let us keep these up — we will return to them at the end of the lesson.' WAIT TIME: after each cold-call, wait 10 seconds before taking the next hand. | prediction to paper before any instruction, activating prior knowledge of cones and nets from Lessons 1–3. Posting predictions publicly creates social accountability and a reference point for later conceptual change. | three sentence-starters completed in their book before group sharing begins. Teacher scans books during the 4-minute window; if fewer than 70 % of learners have written anything, pause and prompt: 'Think about the cone net we drew in Lesson 1 — what did that look like?' Note on the board which groups predict 'more curved surface' vs 'less' — this distribution will be revisited in Phase 3. |
| **Phase 2 — Observe**<br>📹 **VIDEO:**<br>Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Volume of the Cone (PLIX)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Each group receives: one pre-scored paper cone, scissors, a ruler, a compass, a protractor, a piece of A3 squared paper, and a printed Frustum Dimensions Table (columns: R, r, h, l_measured, l_calculated). Step 1 — Cut: one learner per group carefully cuts the cone along the pre-scored line under teacher supervision, producing a frustum (larger piece) and a small cone (smaller piece). Step 2 — Measure: learners measure and record R (base radius of frustum), r (top radius of frustum), and h (vertical height of frustum) in the table. Step 3 — Unfold: learners carefully cut a straight slit up the curved surface of the frustum and lay it flat onto | Before distributing materials, model Steps 1–2 at the front with a demonstration cone. Narrate aloud: 'I am cutting along this line — notice the cut must stay parallel to the base, otherwise R and r will not be true circles.' Circulate during group work with a checklist. At Step 3 stop all groups after 2 minutes and ask whole-class: 'What shape is your unfolded curved surface — is it a full sector, two sectors, or something else?' Cold-call 2 groups. WAIT TIME: 12 seconds after posing the question. At Step 4 prompt: 'Hold your full-cone net from Lesson 1 next to your frustum net. What do you notice about the inner arc that was NOT there before?' Cold-call 3 | Concrete–Representational–Abstract (CRA) sequence — learners move from handling a physical 3-D solid (Concrete) to tracing and labelling its flat net (Representational) before any symbolic formula is introduced (Abstract in Phase 3). This grounds the later algebraic formula in a physical reality learners have personally constructed. | Observable indicators: (1) Every group's Frustum Dimensions Table has numeric entries in all four columns (R, r, h, l_measured) before Phase 3 begins. (2) Every learner's book contains a labelled sketch of the 3-D frustum with R, r, h, and l marked correctly. Teacher checks: if a group's 'net' is not a true annular sector (e.g. the cone was cut at an angle), acknowledge the measurement error and have them re-measure R and r at the actual cut faces rather than re-cutting. Misconception flag: if learners call the two straight edges of the net 'height', intervene: 'That measurement you took directly from the net — is it the vertical height of the frustum, or the |

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| | the A3 squared paper, tracing the outline of the unfolded net. They observe and label: the outer arc (from the large base), the inner arc (from the small top), and the two straight edges (slant height l). They measure l directly from the net. Step 4 — Compare: learners place the frustum net alongside the full cone net drawn in Lesson 1 and record observations about size and shape differences in their books. Step 5 — Record: learners sketch and label the 3-D frustum identifying R, r, h, and l on their diagram. | groups. Write on the board exactly what learners say, verbatim, without evaluating. After Step 5 ask: 'How many radii does a frustum have? Turn and tell your partner.' Cold-call 4 individuals — target at least 1 learner who has not yet spoken today. | | distance along the sloping side? How would you check?' |
| **Phase 3 — Explain** 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Volume of the Cone (PLIX)] Source: CK-12 🔍 Search ARES for similar readings | Learners work in the same groups through a structured Explain sequence. Part A — Deriving the formula conceptually: Using the frustum and the small cone they removed in Phase 2, learners recall (from Lesson 1) that the lateral surface area of a full cone is πrl_cone. They write on mini-whiteboards: Curved SA of frustum = Lateral SA of big cone − Lateral SA of small cone. Teacher guides them to label the big cone's slant height as L, base radius R; and the small cone's slant height as (L − l), base radius r. Learners substitute into the difference, cancel, and arrive at π(R + r)l. Part B — Slant height: Teacher poses a context: 'Suppose you are a metalworker in Gikomba market and you only have a tape measure and a plumb line — you cannot measure l directly. You can measure R, r, and h. How do you find l?' Learners draw a right triangle on their mini-whiteboards: vertical side h, horizontal side (R − r), hypotenuse l. They write and simplify $l = \sqrt{h^2 + (R - r)^2}$. Part C — Kirinyaga Road bucket: | Part A: Write on the board 'Curved SA = πRL − πr(L−l)' with the derivation steps in a table. Ask: 'Before I expand this — what happens to the L terms after we use similar triangles to relate L and l? Discuss for 90 seconds.' Cold-call 2 groups. WAIT TIME: 15 seconds. Complete the derivation step-by-step, pausing at each line to say: 'Agree? Show thumbs up / thumbs down on your mini-whiteboard.' Part B: Do NOT show the right triangle — ask 'Draw me the cross-section of the frustum. What right triangle can you spot inside it?' Cold-call 1 learner to draw on the board. WAIT TIME: 12 seconds. Affirm the correct triangle; narrate: 'The horizontal leg is (R − r) because both radii start from the central axis.' Part C: Say: 'You are the metalworker pricing the Kirinyaga Road bucket. Work out the curved surface area only — silent individual work, 5 minutes.' Circulate, note errors on a tally. After 5 minutes cold-call 3 learners with different answers to write on the board; facilitate peer error-correction. Part D: Ask: 'For | Algebraic derivation with physical referents — learners derive π(R + r)l by subtracting two known cone-surface-area expressions (knowledge from Lesson 1) rather than accepting the formula as given. This 'formula genealogy' strategy ensures the formula is understood, not memorised, and directly answers the driving question 'how do we calculate exactly how much material is needed?' | Observable indicators: (1) In Part A, learner mini-whiteboards show the correct intermediate step πRL − πr(L − l) = πl(R + r) before the teacher confirms it. (2) In Part C, at least 80 % of learners independently arrive at the correct curved SA for the Kirinyaga Road bucket (l ≈ 33.3 cm, curved SA ≈ 3,140 $cm^2$) within the 5-minute window. (3) In Part D, learners correctly identify that total SA = $\pi(R + r)l + \pi R^2$ (closed bottom) before Lesson 6. Misconception to monitor: learners using h instead of l in the formula π(R + r)h — address immediately: 'Point to the slant height on your physical frustum. Is that the same as the vertical height you measured earlier?' |

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| | Learners use R = 18 cm, r = 12 cm, h = 30 cm to calculate l then curved SA. Part D — Eldoret grain silo frustum: Learners use R = 1.5 m, r = 0.6 m, h = 2.4 m; they must also add the area of the two circular ends (large and small base) to find the total surface area, since the silo is open at the top and closed at the bottom. | the Eldoret grain silo, the farmer needs to paint the outside including the base but NOT the open top. What extra areas must we add to the curved surface area?' Cold-call 2 learners. Emphasise connection to phenomenon: 'This is exactly the question we need to answer for the Kirinyaga Road frustum bucket — curved surface PLUS the two circular ends. We will compute the full surface area in Lesson 6.' | | |
| **Phase 4 — Driving Question Board (DQB) Update**<br>📹 **VIDEO:** Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Volume of the Cone (PLIX)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Each group reviews the class DQB (a large chart on the classroom wall tracking the unit's growing understanding). They receive three colour-coded sticky notes: GREEN — 'Something we can now explain'; ORANGE — 'Something that is still not clear'; PINK — 'A new question this lesson raised'. Groups write their notes and one representative from each group posts them on the DQB. Whole class does a silent 2-minute gallery walk around the DQB. Learners individually write one sentence in their books completing the stem: 'Lesson 5 helps me answer the driving question because now I know that…' | Direct attention to the Prediction Wall from Phase 1: 'Earlier, most groups predicted the frustum net would be (more / less) material than the cone. Now that you have computed the curved surface area, were your predictions correct?' Cold-call 3 groups whose sticky notes are still on the wall. WAIT TIME: 10 seconds. Then direct attention to the DQB: 'Look at the ORANGE notes from Lessons 1–4. Which ones can we now move to GREEN?' Facilitate a 3-minute whole-class discussion — do NOT move any note yourself; ask learners to justify before relocating. End by reading aloud two PINK 'new question' notes and asking: 'Which of these do you think Lesson 6 will answer?' Connect explicitly to phenomenon: 'We still have not confirmed whether the shopkeeper on Kirinyaga Road is correct. What are we still missing?' (Answer: the areas of the two circular ends, and the comparison with the cylinder and cone buckets — to be done in Lessons 6–7.) | Collaborative sense-consolidation via DQB — the three-colour protocol forces learners to distinguish between resolved understanding (GREEN), persistent confusion (ORANGE), and new curiosity (PINK). This metacognitive sorting ensures the DQB remains a living tool that tracks conceptual progress rather than a static display. | Observable indicators: (1) Every group posts at least one GREEN note that correctly references either the formula $\pi(R + r)l$ or the relationship $l = \sqrt{h^2 + (R - r)^2}$. (2) Every learner's book contains a completed sentence stem that specifically mentions R, r, or l — not just 'the formula'. (3) ORANGE notes that still mention 'the difference between h and l' signal a group that needs re-teaching at the start of Lesson 6. |
| **Phase 5 — Model Building** | Each learner has a cumulative 'Surface Area Model' diagram in the back of their exercise book — | Project a blank version of the cumulative concept map on the board. Ask: 'Before you add the | Cumulative model revision — by adding each new solid to a single growing concept map, learners | Observable indicators: (1) Every learner's concept map has the frustum node connected to the |

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| 📹 **VIDEO:** [Writing proportional equations from tables] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Volume of the Cone (PLIX)] Source: CK-12 🔍 Search ARES for similar readings | a concept map begun in Lesson 1 showing all solids studied so far (cone, square pyramid, rectangular pyramid, sphere, hemisphere) with their surface-area formulas and key dimensions labelled. In this lesson, learners add the frustum of a cone node. They draw a labelled 3-D sketch of a frustum, write the curved SA formula $\pi(R + r)l$, the total SA formula $\pi(R + r)l + \pi R^2 + \pi r^2$ (for a closed frustum), and the slant-height formula $l = \sqrt{h^2 + (R - r)^2}$. They draw an arrow from the 'cone' node to the 'frustum' node labelled 'slice parallel to base'. They also annotate their model with one real-world Kenyan object for the frustum (e.g. 'tapered metal bucket, Kirinyaga Road'; 'grain silo, Eldoret'; 'lampshade, Nakumatt'). Finally, each learner writes one 'lingering question' at the bottom of their model page to bring to Lesson 6. | frustum, trace the path from cone to frustum on your own map — draw the arrow first, then add the node.' Give 2 minutes of silent individual work. Circulate and check that the arrow is labelled correctly ('slice parallel to base') — this label encodes the geometric definition of a frustum and will be needed when learners encounter frustums of pyramids in Lesson 7. Then say: 'Add the two formulas — curved SA and total SA. Leave the total SA for a closed frustum — we will verify that in Lesson 6 when we add the circle areas.' Ask: 'What Kenyan object in your home or community is shaped like a frustum?' Cold-call 4 learners — target gender balance. WAIT TIME: 10 seconds. Write their suggestions on the board. Close the lesson by returning to the phenomenon: 'The Kirinyaga Road shopkeeper said the frustum bucket uses the most metal. Today you computed the curved surface area. In Lesson 6 we add the bases and compare all three buckets. Keep your Frustum Dimensions Table — you will need it.' | build a connected schema rather than isolated formula lists. The directional arrow 'cone → frustum (slice parallel to base)' encodes the geometric relationship and prepares learners for the pyramid frustum in Lesson 7, making the model genuinely predictive rather than merely descriptive. | cone node with a correctly labelled arrow. (2) The frustum node contains both the curved SA formula and the total SA formula (with $\pi R^2 + \pi r^2$ terms present even if the lesson has not yet verified them numerically). (3) At least one Kenyan real-world object is annotated on the frustum node. (4) A 'lingering question' is written — questions that mention 'the two circle ends' or 'how to compare all three buckets' indicate readiness for Lesson 6; questions that still ask 'why is the formula $\pi(R + r)l$ and not just $\pi Rl$' indicate a learner who needs the Part A derivation revisited at the start of Lesson 6. |

## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Prediction Wall evidence — did the majority of learners predict 'more material' or 'less material' for the frustum compared to the original cone? If most predicted 'less', what does this reveal about their understanding of how removing the apex affects surface area, and how will you address this in Lesson 6's opening? (2) Formula derivation — during Part A of the Explain phase, how many learners were able to write the intermediate step $\pi RL - \pi r(L - l)$ without scaffolding? If fewer than half did, consider providing a partially completed derivation frame at the start of Lesson 6 before asking learners to apply the formula independently. (3) h vs l confusion — did any groups consistently substitute h into the formula instead of l? If yes, plan a targeted 5-minute re-teaching segment at the start of Lesson 6 using the physical frustum: 'Hold a ruler vertically inside your frustum — that is h. Now hold it along the sloping side — that is l. Are they equal?' (4) Kenyan context resonance — which real-world frustum objects did learners suggest during the Model Building phase? Were there surprises (e.g. the base of a kikapu woven basket, the top of a sufuria)? Note these for use in Lesson 6 word problems to increase relevance. (5) DQB quality — review the ORANGE notes posted. Are confusions about slant height, about the number of surfaces to include, or about the relationship between the frustum and the original cone? Use these to sequence the opening of Lesson 6.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
|---|---|
| **What did I observe?** | Learners physically cut a paper cone along a plane parallel to its base to produce a frustum, measured its dimensions (R, r, h, l), unfolded its curved surface onto squared paper, and observed that the net is an annular sector — the region between two concentric arcs — which is larger and more complex than the full cone net from Lesson 1. |
| **What did I learn?** | The curved surface area of a frustum of a cone is given by $\pi(R + r)l$, where R is the radius of the large base, r is the radius of the small top, and l is the slant height. The slant height can be calculated from the vertical height and the two radii using the Pythagorean relationship $l = \sqrt{[h^2 + (R - r)^2]}$. The total surface area of a closed frustum also includes the areas of the two circular ends: $\pi R^2 + \pi r^2$. |
| **How does this explain the phenomenon?** | The formula $\pi(R + r)l$ arises because the curved surface of a frustum is the difference between the lateral surfaces of two similar cones sharing the same apex: Curved SA = $\pi RL - \pi r(L - l) = \pi l(R + r)$. This connects directly to the Kirinyaga Road phenomenon: the tapered (frustum) bucket has two different-sized circular ends and a sloping curved surface whose area depends on both radii simultaneously, which is why its net is unexpectedly large compared to the straight-sided cylindrical bucket or the conical bucket of the same volume. |

# LESSON 6 (80 minutes): Total Surface Area of Frustums of Cones — Open and Closed

## A. SPECIFIC LEARNING OUTCOMES

| Category | Specific Learning Outcomes |
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| **Purpose** | Learners calculate the total surface area of closed and open frustums of cones, and use this to determine how much sheet metal is needed for the tapered bucket in the anchoring phenomenon — resolving whether the shopkeeper on Kirinyaga Road is correct. |
| **Knowledge** | A frustum of a cone is formed by cutting a smaller cone from the top of a larger cone with a parallel cut. Its curved surface area is $\pi(R + r)l$, where R is the larger radius, r is the smaller radius, and l is the slant height. The slant height is found using $l = \sqrt{[h^2 + (R - r)^2]}$. A closed frustum adds both circular faces ($\pi R^2 + \pi r^2$); an open frustum adds only the base ($\pi R^2$) or neither circular face, depending on the physical object. The total surface area formula for a closed frustum is: $TSA = \pi(R + r)l + \pi R^2 + \pi r^2$. |
| **Skills** | Measuring the dimensions of physical frustum-shaped containers (tapered buckets, plastic tumblers) using a ruler and tape measure; substituting values into the frustum surface area formula; computing slant height from perpendicular height and radii; distinguishing between open and closed frustum surface areas; comparing calculated values with manufacturer specifications; interpreting results to resolve the anchoring question. |
| **Attitudes** | Appreciating that the shape of a container — not just its volume — determines the amount of material needed to construct it; developing curiosity and persistence when measurements from physical objects do not immediately match expected formula results; valuing precision in measurement as a life skill relevant to trades such as metalwork, tailoring, and construction. |
| **Key Inquiry Question** | How are solids used in real life? |
| **Purpose in Storyline** | In Lessons 1–5, learners developed surface area tools for cones, pyramids, spheres, and hemispheres. Lesson 6 is the resolution lesson for the anchoring phenomenon: learners now apply the frustum surface area formula to the tapered metal bucket, compute and compare the sheet-metal requirements for all three buckets (cylinder from Lesson 3, cone from Lesson 1, frustum today), and evaluate the shopkeeper's claim directly. This is the evidence-gathering climax of the sequence before Lesson 7 (frustums of pyramids) and Lesson 8 (applications and consolidation). |
| **Safety Notes** | When measuring physical metal or plastic buckets, learners should handle any sharp pressed-metal rims carefully. No cutting tools are used in this lesson. If physical buckets have been washed before class, ensure they are dry to avoid slipping. Remind learners not to stand containers on the edge of desks. |

## B. LESSON OVERVIEW

Learners begin by predicting how much sheet metal the tapered (frustum-shaped) bucket from the Kirinyaga Road hardware shop requires, drawing on their prior knowledge of cones and cylinders. They then derive the frustum curved-surface-area formula by conceptually 'reassembling' the frustum from two cones (connecting to Lesson 1), before measuring real frustum-shaped containers brought from home (tapered buckets, plastic tumblers) and computing both open and closed surface areas. Groups compare their computed values with manufacturer label data where available. In the Explain phase, the class resolves the anchoring question: does the frustum bucket use more sheet metal than the cylindrical or conical bucket of equal 10-litre capacity? Learners update their Driving Question Boards and revise their cumulative net-based models to include the frustum, completing the three-bucket comparison.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Phase 1 — Predict**<br>📹 **VIDEO:**<br>[Local linearity for a multivariable function]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>[Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | Learners are shown a physical tapered bucket (or a clear diagram of the three Kirinyaga Road buckets) and reminded that all three hold exactly 10 litres. Each learner silently writes a personal prediction in their exercise book: 'Which bucket do you think uses the most sheet metal — the cylinder, the cone, or the frustum? Give a reason.' They then share predictions with a partner (Think-Pair-Share) and the class hears 3–4 predictions cold-called. Learners also sketch a rough net of the frustum from memory, labelling any dimensions they think matter. Predictions are recorded on a class 'Prediction Wall' (a section of the board left visible all lesson). | Display the three bucket outlines side by side on the board and say: 'Angalia mifano hii ya ndoo tatu — look at these three bucket shapes. All hold exactly 10 litres. Before we do any mathematics today, I want your honest guess.' Give 90 seconds of silent writing time. After Think-Pair-Share, cold-call exactly 4 learners (choose one who predicted 'frustum uses most', one who predicted 'cylinder uses most', one who predicted 'cone uses most', and one who is unsure). For each, ask: 'What is your reason?' Apply 10-second WAIT TIME before accepting answers. Record each prediction on the board in three columns: Frustum / Cylinder / Cone. Do NOT confirm or deny any prediction — say: 'We will test every single one of these predictions before the lesson ends.' | Think-Pair-Share with a Prediction Wall. This activates prior knowledge from Lessons 1–5 (cone and cylinder surface areas) and creates cognitive dissonance — learners who expect 'same volume = same surface area' will be challenged. The visible Prediction Wall maintains accountability: learners know they will return to these predictions at the end of the lesson. | Circulate during silent writing and scan exercise books. Check: (a) Does the learner reference a formula or dimension (e.g. 'the frustum has a bigger opening so more area')? (b) Does the learner's sketch include R, r, and some height? Learners who draw only a rectangle or a triangle for the net have not yet visualised the frustum's lateral surface — note these learners for targeted support in Phase 2. |
| **Phase 2 — Observe**<br>📹 **VIDEO:**<br>[Local linearity for a multivariable function]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>[Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | Activity A — Deriving the Formula (15 min): In pairs, learners receive a printed diagram showing a frustum alongside the two cones it comes from (a large cone of radius R and slant height L, and the removed small cone of radius r and slant height l_small). Using the curved surface area of a cone formula ($\pi rl$) from Lesson 1, pairs work out: Curved SA of frustum = $\pi RL - \pi r \cdot l_{small}$. Using similar triangles ($R/r = L/l_{small}$), they simplify this to $\pi(R + r)l$, where l is the slant height of the frustum itself. The teacher guides this derivation with structured questions. Pairs also derive $l = \sqrt{h^2 + (R - r)^2}$ from the right-triangle hidden inside the frustum. Activity B — Measuring Real | For Activity A: Write on the board — 'Large cone: radius R, slant height L. Small cone removed: radius r, slant height l_small.' Ask: 'From Lesson 1, what is the curved surface area of a cone?' WAIT 10 seconds. Cold-call 2 learners. Write $\pi RL$ and $\pi \cdot r \cdot l_{small}$ on the board. Say: 'The frustum is what is left after we remove the small cone. So the curved surface area of the frustum is...?' WAIT 12 seconds. Cold-call 1 learner. Guide the class to see $\pi RL - \pi r \cdot l_{small}$. Then say: 'We want one formula using only R, r, and the frustum's own slant height l. Let me show you that similar triangles give us $R/r = L/l_{small}$.' Walk through the algebra slowly, pausing at each step. Ask: 'What | Structured Derivation followed by Measure-and-Compute with Real Objects. The derivation phase builds conceptual understanding by connecting the frustum formula directly to the cone formula learners already know (Lesson 1 anchor), preventing rote memorisation. The hands-on measurement phase grounds the abstract formula in the physical reality of the anchoring phenomenon — learners are literally handling the type of object the shopkeeper sells. | During Activity A, listen for whether pairs correctly identify the subtraction step ($\pi RL - \pi r \cdot l_{small}$) before simplification. A common error is writing $\pi Rl - \pi rl = \pi(R-r)l$ — intercept this immediately by asking: 'Are R and l_small related to the same cone, or different cones?' During Activity B, check that groups measure the perpendicular height h, not the slant side — probe by asking: 'Is your tape measure following the slanted wall of the bucket, or going straight down inside?' Groups whose computed slant height is less than their measured h have made an error — flag this. |

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| | Containers (20 min): Groups of 4 receive one physical frustum-shaped container (tapered bucket, large plastic tumbler, or a lamp shade) and a tape measure plus ruler. Each group measures: the top radius R (or diameter D), the bottom radius r (or diameter d), and the perpendicular height h. They compute slant height l, then calculate: (i) Curved surface area only, (ii) Open frustum TSA (curved + base only), (iii) Closed frustum TSA (curved + both circles). Groups record results in a structured table. Where a manufacturer sticker or label is present, they compare their computed value with the stated surface area or material specification. | does this tell us about L?' WAIT 10 seconds. Cold-call 1 learner. Complete the simplification to $\pi(R+r)l$ on the board. Box the final formula. Then say: 'Now look at the right-angled triangle inside the frustum — the vertical height h, the difference in radii (R−r), and the slant height l. How do we find l?' WAIT 10 seconds. Cold-call 1 learner. Write $l = \sqrt{h^2 + (R-r)^2}$ on the board. For Activity B: Distribute containers and say: 'Measure carefully — these are real objects, just like the buckets at the hardware shop on Kirinyaga Road. Record every measurement in cm to one decimal place.' Circulate. At the 10-minute mark, ask: 'Which group is finding it hardest to measure the diameter? Remember — measure across the widest part, then divide by 2.' After groups finish calculations, ask one group to read out their values and narrate their steps aloud to the class. | | |
| **Phase 3 — Explain**<br>📹 **VIDEO:** [Local linearity for a multivariable function]<br>Source: Khan Academy (English - US curriculum)<br>🔍 [Search ARES for similar videos]<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 [Search ARES for similar readings] | Whole-class resolution of the anchoring question. Each group presents their group's container results (2 minutes per group). The teacher then reveals (or has learners compute) the surface area of the cylindrical bucket (from Lesson 3 class data) and the conical bucket (from Lesson 1 class data), both holding 10 litres. These three values are written side by side on the board. Learners individually answer in writing: 'Was the shopkeeper on Kirinyaga Road correct? Use the numbers on the board to justify your answer.' Learners then work two standard exam-style problems on frustums — one open (a metal bucket with no lid), one closed (a | As groups present, annotate their values on a master comparison table on the board: Bucket Type \| Radius dimensions \| Slant height \| Curved SA \| Total SA. After all groups have presented, say: 'Now look at this table. Remember the shopkeeper's claim: the frustum bucket uses the most metal. Based on our mathematics today — is she or he right?' WAIT 15 seconds (longer, deliberate pause). Cold-call 3 learners with different answers: 'Tell us the numbers you are comparing.' After the discussion, say: 'The answer depends on the exact dimensions of each bucket — the shopkeeper may be correct for the specific buckets in that shop, but it is NOT | Anchoring Phenomenon Resolution with a Three-Way Comparison Table. Displaying all three bucket surface areas simultaneously makes the comparison concrete and visually immediate. The written individual justification ('Was the shopkeeper correct?') is a critical-thinking move that requires learners to use quantitative evidence rather than opinion — directly embodying the CBC competency of critical thinking and problem solving. | Collect the written justifications ('Was the shopkeeper correct?'). Assess for: (1) Does the learner cite specific numerical values from the board? (2) Does the learner acknowledge that the answer depends on dimensions, not just shape? (3) Does the learner correctly identify which formula components are included for open vs. closed frustums? Learners who write only 'yes' or 'no' without numbers have not met the expectation — plan a follow-up probe question for the next lesson. |

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| | painted cylindrical tin with top and bottom). Finally, learners distinguish verbally: 'When is a frustum open at the top only? Open at both ends? Fully closed?' using real Kenyan examples (water bucket — open top; section of a water pipe — open both ends; a sealed tin of Kimbo cooking fat — closed). | automatically true that a frustum always uses more metal than a cylinder of equal volume. Shape and dimensions both matter.' Write this conclusion on the board. For the two exam problems, give 8 minutes of silent individual working time, then pair-share answers. For the open/closed distinction, hold up the Kimbo tin and the plastic bucket and ask: 'Which surfaces do we include for each? Why?' WAIT 10 seconds between each object. Cold-call 2 learners per object. | | |
| **Phase 4 — Driving Question Board (DQB) Update**<br>📹 **VIDEO:** Local linearity for a multivariable function<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Each learner retrieves their personal DQB card (or the class DQB is updated collectively). Learners add one green sticky note (or written entry) stating something they can now answer about the driving question, and one orange sticky note for something they are still wondering. Suggested prompts on the board: GREEN — 'I can now explain why the frustum bucket costs more/less because...'; ORANGE — 'I am still wondering whether...'. The class briefly shares 2–3 green and 2–3 orange contributions. The teacher records any remaining orange questions that will be addressed in Lesson 7 (frustums of pyramids) or Lesson 8 (applications). | Say: 'Take out your DQB cards. Our driving question is: How do we calculate exactly how much material is needed to cover or construct any solid shape — and why does the shape of a container matter even when the volume stays the same? We just answered the second half of that question today for the frustum. Add your green and orange notes now.' Give 3 minutes of silent writing. Then cold-call 2 green entries and 2 orange entries. For each orange, say: 'Good question — is that something we have already started to answer, or something new?' Physically move unresolved orange notes to the 'Still to Investigate' column of the class DQB. Point to it and say: 'These are the questions that will drive Lessons 7 and 8.' | Driving Question Board with Colour-Coded Evidence Tracking. The green/orange distinction makes visible to learners (and teacher) exactly where understanding has grown and where gaps remain. Returning to the same driving question across all 8 lessons builds the metacognitive habit of tracking cumulative understanding — a core CBC learning-to-learn competency. | Read all green sticky notes/entries quickly after class. Check: Are learners naming the correct formula and connecting it to the phenomenon (bucket/metal/shopkeeper), or writing generic statements ('I learned surface area')? Generic statements indicate surface-level engagement. Count how many orange notes reference misconceptions from Phase 1 predictions that were not resolved — these learners need the worked example at the start of Lesson 7. |
| **Phase 5 — Model Building**<br>📹 **VIDEO:** Local linearity for a multivariable function<br>Source: Khan | Learners retrieve their cumulative net-based model (a large sheet of paper on which they have been building nets of all solids studied since Lesson 1). Today they add the net of a frustum: two concentric annular (ring-shaped) sectors for the curved surface, | Say: 'Open your cumulative model sheet. Find the space we reserved for the frustum. You are going to draw the net today.' Draw the net on the board step by step: first the large annular sector (outer arc radius = L = slant height of large cone, inner arc radius = l_small), | Cumulative Net-Based Model Revision. The running model sheet, built across all 8 lessons, functions as a physical knowledge-organisation tool — equivalent to a science notebook's running model in NGSS practice. Annotating the frustum net with real values from | Collect or photograph cumulative model sheets at the end of class. Check for: (1) Correct net shape (annular sector, not a flat trapezium); (2) All four labels present: R, r, h, l; (3) Formula written correctly with π(R+r)l as the curved part; (4) Phenomenon |

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| Academy (English - US curriculum) 🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings | plus two circles of radius R and r for the closed version. Learners label all dimensions: R, r, h, l, and write the formula TSA = $\pi(R+r)l + \pi R^2 + \pi r^2$ directly on their model next to the frustum net. They then annotate which surfaces are removed for an open frustum (cross out the top circle). Finally, learners write one sentence on their model connecting the frustum back to the anchoring phenomenon: e.g. 'This net represents the tapered metal bucket on Kirinyaga Road. The shopkeeper uses this much metal per bucket: [value from class data].' | then the smaller sector removed, leaving the frustum's lateral net. Say: 'This curved shape, when rolled up, makes the side of your tapered bucket.' WAIT and let learners draw. Then say: 'Add both circles — one for the base, radius R; one for the top, radius r. If your bucket is open at the top, cross out the top circle and write: open frustum.' Circulate and check that dimensions are correctly labelled. In the last 2 minutes, ask 2 learners to share their model annotation sentence about the phenomenon. Say: 'Next lesson we move to frustums of pyramids — the shape you might see in a Maasai granary or a rooftop. Keep your model ready.' | the Kirinyaga Road phenomenon bridges abstract geometry and real-world application, reinforcing the sub-strand's key inquiry: 'How are solids used in real life?' | connection sentence is specific (includes a number). Learners whose nets show a flat trapezium have a persistent misconception about how the curved lateral surface unrolls — address this individually at the start of Lesson 7. |

| D. TEACHER REFLECTION |
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| After the lesson, reflect on the following: (1) Resolution quality — Did most learners correctly compare the three bucket surface areas using numbers, or did they rely on intuition? If more than one-third gave a non-numerical justification in the written response, plan a 10-minute recap at the start of Lesson 7 using the three values from today's comparison table. (2) Measurement accuracy — Which groups had the largest discrepancy between their computed value and the manufacturer label (or teacher's reference value)? What was the likely source of error — measuring slant height instead of perpendicular height, or using diameter instead of radius? Name this error explicitly at the start of Lesson 7. (3) Formula derivation — Were learners able to follow the similar-triangles step, or did most simply accept the formula? If the derivation felt rushed, consider providing a printed step-by-step derivation card as a reference tool for Lesson 7. (4) Open vs. closed distinction — Could learners correctly identify which real-world objects are open, partially open, or closed frustums? If confusion persisted, add two more Kenyan examples to the Lesson 7 warm-up (e.g. a chapati-rolling metal ring — open both ends; a sealed uji thermos — closed). (5) DQB orange notes — How many orange notes referenced questions that go beyond the sub-strand (e.g. 'What if the bucket is not a perfect frustum?')? Use these to build extension tasks for fast finishers in Lesson 8. |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
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| **What did I observe?** | Learners measured real frustum-shaped containers (tapered buckets, plastic tumblers) brought from home, recording the top radius R, bottom radius r, and perpendicular height h. They compared computed surface areas across groups and, where available, checked against manufacturer label data. They also observed the derived net of a frustum — two concentric annular sectors plus two circles — which is visibly larger and more complex than the rectangular net of a cylinder of equal height. |
| **What did I learn?** | The curved surface area of a frustum of a cone is $\pi(R + r)l$, where $l = \sqrt{[h^2 + (R - r)^2]}$ is the slant height. A closed frustum's total surface area adds both circular faces: TSA = $\pi(R + r)l + \pi R^2 + \pi r^2$. An open frustum (such as a metal bucket with no lid) uses only TSA = $\pi(R + r)l + \pi R^2$. Whether the frustum uses more sheet metal than a cylinder or cone of equal volume depends on the specific dimensions of each shape — the shopkeeper's claim may hold for those particular buckets, but it is not a universal rule. |

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| **How does this explain the phenomenon?** | Because the frustum is formed by removing a small cone from a large one, its lateral surface unfolds into a large annular (ring-shaped) sector rather than a simple rectangle or triangle. The formula $\pi(R + r)l$ captures the fact that both the large and small circular edges contribute to the boundary of this curved surface. Adding the areas of the one or two circular faces to the curved surface area gives the total amount of sheet metal needed — which is why a tapered bucket can require a different (and often larger) sheet of metal than a straight-sided cylindrical bucket of identical volume. |

| LESSON 7 (80 minutes): Surface Area of Frustums of Pyramids | |
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| **A. SPECIFIC LEARNING OUTCOMES** | |
| **Category** | **Specific Learning Outcomes** |
| **Purpose** | Learners extend their frustum understanding from circular cross-sections (Lesson 6) to pyramidal frustums with square or rectangular bases, deriving a formula by summing the areas of all faces and verifying it against a physical cardboard net. |
| **Knowledge** | Learners will know that a frustum of a pyramid has two parallel rectangular/square bases and four trapezoidal lateral faces, and that its total surface area is the sum of the areas of both bases plus the sum of the areas of all trapezoidal lateral faces. |
| **Skills** | Learners will be able to: (i) identify and label all faces of a pyramidal frustum; (ii) derive the surface area formula by summing component face areas; (iii) calculate the slant height of each trapezoidal face; (iv) apply the formula to real-life objects such as a planter box or trophy stand base; (v) build a cardboard net and verify computed surface area against direct measurement. |
| **Attitudes** | Learners appreciate that geometric reasoning — not guesswork — is the reliable tool for answering the shopkeeper's claim, and they develop persistence when handling the multi-step derivation of the pyramidal frustum formula. |
| **Key Inquiry Question** | How are solids used in real life? |
| **Purpose in Storyline** | In Lessons 1–6 learners computed surface areas of cones, pyramids, spheres, hemispheres, and conical frustums. The shopkeeper's claim introduced in the anchoring phenomenon has been partially tested. This lesson equips learners with the final missing tool — the surface area of a pyramidal frustum — so that in Lesson 8 they can fully compare all three bucket types and deliver a mathematically justified verdict on whether the frustum truly uses the most sheet metal. |
| **Safety Notes** | When cutting cardboard with scissors or craft knives, learners must cut away from the body and keep fingers clear of blade paths. Rulers used as cutting guides must be held firmly. Scrap cardboard and paper offcuts must be collected and disposed of responsibly to keep the classroom tidy and safe. |

## B. LESSON OVERVIEW

Learners begin by predicting whether a pyramidal frustum (square-based) will have more or fewer faces than a conical frustum, then examine physical cardboard models to observe all six faces (two bases + four trapezoids). They measure and record dimensions, derive the surface area formula step by step, and apply it to a realistic Kenyan context — the decorative base of a trophy stand used at a Nairobi secondary school inter-house athletics competition. They then build their own cardboard net, compute the surface area, and verify by direct measurement. The lesson closes with a DQB update and cumulative model revision, bringing learners one step closer to settling the shopkeeper's claim.

## C. LESSON IMPLEMENTATION FRAMEWORK

| Phase | Learner Experience | Teacher Moves | Sensemaking Strategy | Formative Assessment Strategy |
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| **Phase 1 — Predict**<br>📹 **VIDEO:** [VSEPR for 4 | Learners are shown a photograph of two objects side by side on the board: a conical frustum (the metal bucket from the anchoring | Display the two images clearly (printed A3 sheet or projected). Read the prompt aloud slowly. Say: 'I want every learner writing | Prediction Journaling — by committing a prediction to writing before instruction, learners activate prior knowledge about | Scan exercise books during the 3-minute writing period. Look for: (i) whether learners identify 6 faces (correct) vs. 5 or fewer (reveals |

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| electron clouds<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | phenomenon) and a pyramidal frustum (a square-based trophy stand pedestal sold at a hardware shop on Kirinyaga Road). The teacher poses the written prompt: 'The conical frustum bucket and the pyramidal frustum pedestal both taper from a wider top to a narrower bottom. Without calculating anything, predict: (a) How many flat faces does the pyramidal frustum have? (b) Do you think it will be harder or easier to calculate the surface area of the pyramidal frustum compared to the conical frustum? Write your prediction and one reason in your exercise books.' Learners write individually for 3 minutes, then share predictions with a seat-partner for 1 minute. | — this is your personal prediction, there are no wrong answers yet.' Allow 3 minutes of silent individual writing. Set a visible timer. After 3 minutes say: 'Share your prediction with your neighbour — you have 60 seconds.' After sharing, cold-call 4 learners (choose 2 who look confident and 2 who look uncertain): 'Achieng, how many faces did you predict?' → WAIT 12 seconds. Record all predicted face counts on the board in a tally without correcting. Then ask: 'Kamau, why did you say it would be harder or easier?' → WAIT 12 seconds. Do not confirm or deny. Close by saying: 'Keep your predictions visible — we will check them very soon.' | faces of 3-D solids (built across Lessons 1–6) and create a cognitive hook that sharpens attention during the observe phase. Recording predictions publicly creates a shared reference point the class will revisit. | confusion with pyramid or cone); (ii) whether reasoning references 'trapezoidal faces' or 'two bases' — indicating transfer from the conical frustum lesson. Note names of learners who draw a sketch alongside their prediction — these learners demonstrate strong spatial reasoning and can be called on during the observe phase. |
| **Phase 2 — Observe**<br>📹 **VIDEO:** VSEPR for 4 electron clouds<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Each group of 4 receives: one pre-cut cardboard pyramidal frustum model (assembled), one flat unfolded net of the same frustum, a ruler, and a coloured marker. Task card reads: 'STEP 1 — Handle the solid model. Count all faces and write their shapes in the table below. STEP 2 — Unfold the net and lay it flat. Use your marker to label each face: Top Base (T), Bottom Base (B), Face 1, Face 2, Face 3, Face 4. STEP 3 — Measure and record in millimetres: (a) side length of the top square, a; (b) side length of the bottom square, b; (c) the slant height of one trapezoidal face, l (the slant distance along the face from the midpoint of a bottom edge to the midpoint of the corresponding top edge). STEP 4 — Sketch one trapezoidal face and label its parallel sides and its height.' Learners work in groups for 12 minutes. Groups then do a gallery | Distribute materials before the phase begins to avoid wasted time. Circulate continuously. At the 4-minute mark, crouch beside any group still counting faces and say: 'Run your finger along every face — every flat surface counts. How many flat surfaces can your finger land on?' → WAIT 12 seconds. At the 8-minute mark, ask the whole class: 'Raise your hand if your group has labelled all four trapezoidal faces with different colours.' Acknowledge. Then ask one group: 'Wanjiku, what is the shape of each lateral face?' → WAIT 12 seconds. Expected answer: trapezoid/trapezium. If a group measures the slant edge (corner-to-corner) instead of the slant height (midpoint-to-midpoint), redirect: 'The slant height goes from the middle of the bottom edge straight up to the middle of the top edge — like the slant height on a cone. Re-measure.' During gallery | Concrete-to-Abstract Net Analysis — learners physically handle and unfold the solid before any formula is introduced. This grounds the abstract formula in observable geometric reality: each term in the formula will correspond to a face they have already touched and measured. The gallery walk surfaces differences in measurement technique (slant height vs. slant edge) before those errors propagate into calculations. | Collect one net per group at the end of the phase. Check: (i) all six faces correctly labelled — indicates readiness for derivation; (ii) slant height measured from midpoint to midpoint on the trapezoidal face — if incorrect, this is a targeted intervention point in Phase 3; (iii) the sketch of one trapezoidal face with both parallel sides (a and b) and the perpendicular height labelled — indicates the learner can connect the 2-D shape to the 3-D face. |

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| | walk (3 minutes) to see how other groups labelled their nets. | walk, prompt learners: 'Look at how Group 3 labelled their slant height — is it in the same position as yours?' | | |
| **Phase 3 — Explain**<br>📹 **VIDEO:** [VSEPR for 4 electron clouds]<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)]<br>Source: CK-12<br>🔍 Search ARES for similar readings | The teacher leads a whole-class guided derivation on the board, with learners following along in their exercise books. The derivation proceeds in four visible steps: STEP 1 — 'We agree the pyramidal frustum has 6 faces. Name them.' (Learners call out.) STEP 2 — 'Area of the bottom square base = $b^2$. Area of the top square base = $a^2$. Write that.' STEP 3 — 'Each trapezoidal face has parallel sides a and b, and slant height l. Area of one trapezoid = ½(a + b) × l. There are 4 such faces. So lateral surface area = 4 × ½(a + b) × l = 2(a + b)l.' STEP 4 — 'Total Surface Area = $a^2 + b^2 + 2(a + b)l$.' Learners then work through a contextual problem in pairs: 'A trophy stand pedestal used at the Nakuru County inter-school athletics championship is in the shape of an open-top pyramidal frustum (no top base). The bottom square has side 30 cm, the top square opening has side 18 cm, and the slant height of each trapezoidal face is 14 cm. Calculate the area of sheet metal used to make it.' After 8 minutes of pair work, one pair presents on the board. The teacher closes by asking: 'How does this formula compare to what we found for the conical frustum last lesson?' (Compare: conical used π(R+r)l; pyramidal uses 2(a+b)l for lateral area — both are 'slant height × perimeter-related term'.) | Write each step as a fraction-free equation in large clear numerals. At STEP 3, pause and cold-call 3 learners before writing: 'Before I write, Omondi — what is the formula for the area of a trapezoid?' → WAIT 13 seconds. 'Fatuma — what are the two parallel sides in our context?' → WAIT 12 seconds. 'Kiprono — if one trapezoid has area ½(a+b)l, and there are four identical ones, what is the total lateral area?' → WAIT 14 seconds. During pair work, circulate and listen for the error of including a top base (the pedestal is open-top). If heard, crouch and say quietly: 'Read the problem again — is there a top face on this pedestal?' After the pair presents, ask the class: 'Does everyone agree with this answer? Give a thumbs up, sideways, or down.' Address thumbs-down learners specifically. For the closing comparison question, cold-call 2 learners and say: 'In one sentence, what is similar between the two frustum lateral area formulas?' → WAIT 12 seconds. | Step-by-Step Derivation with Annotated Formula Building — rather than presenting the formula as a given, the teacher builds it live in front of learners from components they have already observed and measured. This 'formula archaeology' approach ensures learners understand why each term exists. The contextual problem (trophy stand at a Nakuru athletics championship) immediately applies the formula to a Kenya-relevant scenario, reinforcing the lesson's connection to the driving question about manufacturing real objects. | While circulating during pair work, use a mental checklist: (i) Does the learner correctly exclude the top base area for the open-top pedestal? (ii) Does the learner correctly substitute a = 18, b = 30, l = 14? (iii) Is the arithmetic carried out correctly: 2(18+30)(14) = 2(48)(14) = 1344 cm², plus bottom base $30^2$ = 900 cm², total = 2244 cm²? Learners who arrive at 2244 cm² meet expectations. Learners who also spontaneously note the units (cm²) and check their answer by estimating exceed expectations. Learners who include the top base area (getting 2568 cm²) need targeted follow-up on reading problem constraints. |
| **Phase 4 — Driving** | The teacher directs learners' attention to the class Driving | Give learners 4 minutes to write both sentences silently. Insist on | Metacognitive DQB Mapping — the two-column DQB structure | Read all posted 'We now know…' notes. Look for: (i) Correct |

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| **Question Board (DQB) Creation** 📹 **VIDEO:** [VSEPR for 4 electron clouds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings | Question Board, which carries the question: 'How do we calculate exactly how much material is needed to cover or construct any solid shape — and why does the shape of a container matter even when the volume stays the same?' Each learner writes on a sticky note (or paper strip) one sentence beginning with either 'We now know…' or 'We can now calculate…' and one sentence beginning with 'We still need to figure out…'. Learners post their notes in two columns on the DQB: KNOW and STILL NEED. The teacher reads 4–5 notes aloud and briefly narrates the lesson's contribution to the sequence: 'In Lessons 1 through 7, we have built up every tool we need. In Lesson 8, we bring it all together to answer the shopkeeper's claim about the three buckets on Kirinyaga Road.' | the sentence starters — they force precision. As learners post notes, scan quickly for misconceptions. Read aloud at least one 'We still need to…' note that correctly identifies the remaining gap (e.g., 'We still need to compare all three buckets using the same volume'). Say: 'This learner has identified exactly what Lesson 8 will do — well done.' Point to the accumulated DQB and say: 'Look at how this board has grown from Lesson 1. Every solid shape we have studied is represented here. The shopkeeper on Kirinyaga Road has no idea what is coming.' This narrative moment builds anticipation and reinforces the storyline coherence. | (KNOW / STILL NEED) makes the boundary of learners' current understanding explicit and public. By naming what remains unknown, learners develop scientific epistemic humility and maintain curiosity about the unresolved driving question. This strategy also gives the teacher real-time diagnostic information about collective readiness for the culminating Lesson 8. | identification of the formula SA = $a^2 + b^2 + 2(a+b)l$ — indicates lesson objective met; (ii) Spontaneous connection to previous lessons (e.g., 'We now know frustums of both cones and pyramids, so we can compare them') — indicates integration across the unit; (iii) 'We still need…' notes that identify the comparison task — indicates readiness for Lesson 8. Flag any 'We now know…' notes that state an incorrect formula for targeted one-on-one correction before Lesson 8. |
| **Phase 5 — Model Building** 📹 **VIDEO:** [VSEPR for 4 electron clouds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos<br><br>📖 **HTML:** [Surface Area and Nets (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings | Each learner individually builds a cardboard net of a pyramidal frustum using the dimensions from the Phase 3 trophy stand problem (bottom square 30 cm, top square 18 cm, trapezoidal slant height 14 cm). Guided steps on a task card: '1. Draw and cut out the bottom square base (30 cm × 30 cm). 2. Draw and cut out the top square (18 cm × 18 cm). 3. Draw and cut out four congruent trapezoids (parallel sides 30 cm and 18 cm, slant height 14 cm). 4. Assemble by taping. 5. Lay the net flat and use a ruler to measure the total area of all faces by direct measurement. 6. Compare your measured total to your calculated value of 2244 cm². Record the difference and suggest one reason for any discrepancy.' After completing the model, learners | Before learners begin cutting, hold up a completed teacher model and say: 'Notice that the four trapezoidal faces must fit exactly around the two square bases when folded — if your trapezoids are wrong, the solid will not close. Measure twice, cut once.' Circulate during construction. At the midpoint (after cutting, before assembly), pause the class and say: 'Hold up your four trapezoids. Check with your neighbour: do they look congruent? Are the parallel sides clearly 30 cm and 18 cm?' → WAIT 10 seconds. Cold-call 2 learners: 'Njoroge, what is the perpendicular height of your trapezoid — not the slant height, the vertical height inside the trapezoid shape?' → WAIT 13 seconds (this requires Pythagoras or direct measurement — it | Physical Model Verification — learners test their formula-derived answer against empirical measurement of their own constructed net. This closing loop between computation and physical evidence mirrors scientific practice (constructing and using models; using mathematics and computational thinking). The journal annotation integrates the new learning into the cumulative conceptual model of surface area of solids built across all 7 lessons, making the progression of understanding visible to both learner and teacher. | Collect the cumulative journal at the end of the lesson. Assess: (i) Is the net geometrically correct — do all six faces have correct dimensions? (ii) Is the computed value (2244 cm²) present and correct? (iii) Is the discrepancy between measured and computed area within a reasonable tolerance (±5%) and is a plausible reason given (e.g., paper thickness, measurement rounding)? (iv) Does the comparison sentence in the journal show understanding of the structural difference between the conical and pyramidal frustum nets? Learners who correctly address all four points exceed expectations for this lesson. |

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| | add a labelled diagram to their cumulative 'Solid Shapes Model Journal' — a running record they have maintained since Lesson 1 — and annotate it with the formula and the comparison to the conical frustum from Lesson 6. | distinguishes slant height from vertical height and deepens understanding). During journal annotation, prompt: 'Write one sentence comparing your pyramidal frustum net to the conical frustum net from last lesson — which was harder to construct and why?' | | |

| D. TEACHER REFLECTION |
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| After the lesson, reflect on the following questions: (1) Did the transition from the conical frustum (Lesson 6) to the pyramidal frustum feel smooth for learners, or did the shift from curved to flat lateral faces cause confusion? What would you do differently to bridge that transition? (2) How accurately did learners measure the slant height of the trapezoidal face — did the common error of measuring the slant edge (corner to corner) persist despite your redirection during Phase 2? (3) During the trophy stand problem, how many learners incorrectly included the top base area? Is this a reading-comprehension issue or a geometric understanding issue, and how will you address it before Lesson 8? (4) Looking at the DQB 'We still need…' sticky notes: do learners clearly understand that Lesson 8 is a comparison lesson, or do some think there is still new content to learn? (5) Were the cardboard net construction times realistic — did any group fall significantly behind, and if so, should the net dimensions be pre-scored before the lesson to save time? |

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
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| **What did I observe?** | Learners handled physical cardboard models and unfolded nets of a pyramidal frustum, counted and labelled all six faces (two square bases and four trapezoidal lateral faces), and measured the bottom side length ($b$), top side length ($a$), and slant height ($l$) of each trapezoidal face. |
| **What did I learn?** | The total surface area of a closed pyramidal frustum with square bases is $SA = a^2 + b^2 + 2(a+b)l$, where $a$ is the top side length, $b$ is the bottom side length, and $l$ is the slant height of each trapezoidal face. For an open-top frustum, the top base area ($a^2$) is omitted. |
| **How does this explain the phenomenon?** | Learners explained why the formula has three distinct parts — two base areas and a combined lateral area — by connecting each algebraic term to a physically observed and measured face on their cardboard net. They applied this explanation to calculate the sheet metal area of a trophy stand pedestal used at a Nakuru County athletics championship, and verified their result by directly measuring the area of their assembled net. |

## LESSON 8 (80 minutes): Consolidation, Applications and Model Revision: Comparing All Three Buckets and Finalising the Surface Area Model

### A. SPECIFIC LEARNING OUTCOMES

| Category | Specific Learning Outcomes |
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| **Purpose** | To consolidate understanding of surface area of all solids studied — cones, pyramids, spheres, hemispheres, and frustums — by applying computational evidence to the anchoring phenomenon, comparing the three buckets of equal volume, and completing a portfolio task on a real-life Kenyan object. |
| **Knowledge** | The learner can recall and apply the surface area formulae for cones, pyramids (square and rectangular base), spheres, hemispheres, and frustums of cones and pyramids. The learner understands that equal volumes do not imply equal surface areas, and that the frustum's additional lateral complexity (two radii plus slant height) typically generates greater surface area than a simple cylinder or open cone of equal volume. |
| **Skills** | The learner is able to: (i) perform multi-step surface area calculations for all five solid types, (ii) compare surface areas computationally to evaluate a real-world claim, (iii) select an appropriate Kenyan real-life object and independently compute its surface area, (iv) present mathematical arguments to peers using evidence, and (v) revise a cumulative surface area model to reflect all solids studied across the unit. |
| **Attitudes** | The learner appreciates the utility of mathematics in everyday commercial and engineering decisions; values precision and thoroughness in multi-step calculation; demonstrates confidence in presenting mathematical justifications; and appreciates the diversity of solid shapes used in Kenyan design and industry. |
| **Key Inquiry Question** | How are solids used in real life? |
| **Purpose in Storyline** | This is the final lesson of the unit. Learners return to the hardware shop on Kirinyaga Road with a full mathematical toolkit. They calculate the surface areas of all three buckets — cylindrical, conical, and frustum-shaped — holding exactly 10 litres, and use computational evidence to evaluate the shopkeeper's claim. Group presentations, a portfolio task on a chosen Kenyan solid, DQB finalisation, and cumulative model revision complete the storyline. |
| **Safety Notes** | No chemicals or sharp tools are used in this lesson. If physical bucket models or nets are handled, learners should take care with any metal-edged cut-outs. Ensure group presentations are conducted in a respectful, orderly manner. Online safety guidelines apply if learners use digital devices to search for real-life dimensions of Kenyan structures. |

### B. LESSON OVERVIEW

Lesson 8 is the culminating lesson of Sub-Strand 2.6. Learners open by revisiting their Lesson 1 predictions about which bucket uses the most metal, then work in groups to carry out full surface area calculations for the cylindrical, conical, and frustum-shaped buckets — all holding 10 litres (volume = 10 000 $cm^3$). Groups present their findings and use mathematical evidence to adjudicate the shopkeeper's claim. In the second half, each learner selects one real-life Kenyan solid (conical grain store, frustum water tank, spherical water tank, etc.) and completes a structured portfolio task. The lesson closes with DQB finalisation — every question card is revisited and marked resolved or extended — and a whole-class revision of the cumulative surface area model, contrasting the skeletal model from Lesson 1 with the fully annotated final version. Formative assessment is gathered through group presentations, portfolio work, and a short exit task.

| C. LESSON IMPLEMENTATION FRAMEWORK | | | | |
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| **Phase** | **Learner Experience** | **Teacher Moves** | **Sensemaking Strategy** | **Formative Assessment Strategy** |
| **Phase 1 — Predict: Revisiting the Original Predictions**<br>📹 **VIDEO:** Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Learners retrieve their Lesson 1 prediction cards (or exercise-book entries) where they first guessed which of the three buckets — cylindrical, conical, or frustum-shaped — would use the most metal. Working individually for 3 minutes, they re-read their original prediction and write a one-sentence 'then vs. now' statement: what they thought in Lesson 1 and what they now believe, based on evidence gathered across the unit. Pairs then share their statements with a neighbour before selected learners share with the class. | Display the three bucket diagrams on the board (or hold up the physical models used in Lesson 1). Say: 'Cast your mind back to Lesson 1 — you made a prediction about which bucket uses the most metal. Retrieve that prediction now.' Allow 3 minutes of silent individual reflection. Then say: 'Write one sentence: what did I think then, and what do I think now?' Allow 2 minutes of writing. Cold-call 4 learners using name sticks — target at least one learner who originally predicted the cylinder and one who predicted the cone, to surface the range of prior thinking. After each share, ask: 'What evidence changed your mind?' Wait 12 seconds before accepting a response. Do NOT confirm or correct yet — build anticipation for the calculation phase. | Predict–Revisit Routine: By explicitly comparing an earlier prediction to current thinking, learners activate prior knowledge from all seven lessons and create a felt need to resolve the question with calculation. This strategy surfaces misconceptions still present (e.g., learners who still believe equal volume implies equal surface area) so the teacher can address them during group work. | Listen for learners who still conflate volume and surface area (e.g., 'they hold the same so they must use the same metal'). Note these learners by name — direct them to a targeted prompt card during Phase 2 group work: 'Sketch the net of a frustum and a cylinder of the same height. Which net has more pieces? Why?' Observable indicator of readiness: learner can articulate at least one geometric reason why shape affects surface area independently of volume. |
| **Phase 2 — Observe: Group Calculation and Evidence Gathering**<br>📹 **VIDEO:** Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook) | Learners work in groups of four. Each group receives a 'Bucket Data Sheet' giving the pre-calculated dimensions for three open-topped buckets, each holding exactly 10 litres (10 000 $cm^3$): (A) Cylinder — radius r = 10.6 cm, height h = 28.4 cm; (B) Open Cone (apex at bottom, open top) — base radius r = 13.4 cm, slant height l = 29.3 cm; (C) Frustum — top radius R = 14 cm, bottom radius r = 8 cm, slant height l = 24.1 cm, with perpendicular height h = 23.4 cm. Groups calculate the lateral surface area plus base area for each bucket using the formulae established across Lessons 2–7: | Distribute Bucket Data Sheets and large sugar-paper sheets. Say: 'You now have every formula you need from Lessons 2 to 7. Your job today is to be the mathematician the shopkeeper should have hired. Calculate the surface area of each bucket, show every step, and decide: is the shopkeeper correct?' Circulate continuously. At the Cylinder group calculations, pause and ask: 'Why are we adding $\pi r^2$ here? What part of the real bucket does that represent?' (the metal base — wait 10 seconds). At the Frustum groups, ask: 'What does the term $\pi(R+r)$ capture geometrically?' (the average circumference of the two | Collaborative Calculation with Gallery Display (NGSS Science and Engineering Practice 5 — Using Mathematics and Computational Thinking): Groups use mathematics as a tool for evidence generation, mirroring how engineers at a metal-fabrication workshop along Industrial Area, Nairobi, would cost sheet-metal usage. Displaying all group sheets simultaneously allows cross-group comparison and natural error-checking — learners notice if one group's frustum area is dramatically different and investigate why. | Circulate and check: (i) Are learners correctly identifying which formula applies to which bucket? (ii) Are they substituting dimensions accurately (common error: using diameter instead of radius)? (iii) Is the frustum formula including both $\pi(R+r)l$ AND the base $\pi r^2$? Mark a ✓, △, or ? on each group's sheet as you pass. Observable indicator of mastery: group correctly computes frustum lateral area ≈ $\pi(14+8)(24.1)$ ≈ 1667 $cm^2$, base ≈ $\pi(8^2)$ ≈ 201 $cm^2$, total ≈ 1868 $cm^2$; cylinder total ≈ $2\pi(10.6)(28.4) + \pi(10.6^2)$ ≈ 1890 + 353 ≈ 2243 $cm^2$; cone total ≈ $\pi(13.4)(29.3) + \pi(13.4^2)$ ≈ 1232 + 564 ≈ 1796 $cm^2$. [Note: under |

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| Source: CK-12<br>🔍 Search ARES for similar readings | Cylinder: $A = \pi r l_{total} = 2\pi rh + \pi r^2$; Cone: $A = \pi rl + \pi r^2$; Frustum: $A = \pi(R+r)l + \pi r^2$ (open top, closed base). Groups record all working on large sugar-paper sheets to be displayed. Each group also annotates their sheet with a circled verdict: 'The shopkeeper is CORRECT / INCORRECT because…' | circular ends — wait 12 seconds). If a group finishes early, extend: 'Now calculate the surface area if the frustum also had a lid — how does that change the comparison?' Cold-call one reporter per group at the end of this phase (not necessarily the strongest mathematician — use name sticks). Post all sugar-paper sheets on the classroom wall as a 'gallery'. | | these specific equal-volume constraints the open cylinder uses the most metal — this is the productive surprise that enriches the lesson.] |
| **Phase 3 — Explain: Group Presentations and Final Explanation**<br>📹 **VIDEO:**<br>Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:**<br>Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | Each group nominates a spokesperson to present their sugar-paper findings in a 3-minute 'evidence presentation' to the class. Presenters must: (a) state their calculated surface areas for all three buckets, (b) identify which bucket uses the most metal, and (c) use one sentence to agree or disagree with the shopkeeper, citing their numbers as evidence. After all groups present, the class participates in a structured 'Evidence Court' discussion: the teacher poses the verdict question and learners vote with hands, then the teacher facilitates resolution. Learners then individually write a 'Final Explanation' paragraph in their exercise books: 'Based on our calculations, the bucket that uses the most metal is ___ because ___. The shopkeeper's claim is [correct/incorrect/partially correct] because ___. This shows that equal volume does NOT mean equal surface area because ___.' | Facilitate presentations using a visible 3-minute timer. After each group, ask the audience: 'Does this group's answer agree with yours? If not, where do the numbers differ?' Wait 10 seconds. After all groups present, conduct the Evidence Court: 'Hands up — who calculated that the frustum uses the most metal? Who found the cylinder? Who found the cone?' Tally on the board. Say: 'Let us adjudicate — the numbers are the judge, not opinion.' Walk the class through the correct calculation for each bucket on the board, narrating each step: 'For the open frustum, we need the curved surface plus the base — we do NOT include the top because it is open. So our formula is $\pi(R+r)l + \pi r^2$.' After reaching the verdict (under these dimensions, the open cylinder uses the most metal), say: 'So is the shopkeeper always wrong? Or does it depend on the specific dimensions?' Wait 15 seconds — this is a rich discussion point. Prompt: 'What if the frustum had a steeper taper? Would that change things?' After discussion, give 5 minutes for individual Final Explanation writing. Cold-call 3 learners to read their paragraphs aloud — prioritise | Evidence Court Discussion followed by Individual Written Explanation (NGSS Practice 7 — Engaging in Argument from Evidence; NGSS Practice 6 — Constructing Explanations): The public presentation ritual forces learners to articulate mathematical reasoning for an audience, strengthening both communication and conceptual clarity. The individual writing step ensures every learner personally constructs the explanation rather than free-riding on group work. Connecting back to the shopkeeper's claim on Kirinyaga Road grounds the abstraction in a memorable Kenyan commercial context. | Collect and scan the Final Explanation paragraphs. Look for three quality indicators: (1) correct identification of the highest-surface-area bucket with numerical support; (2) explicit statement that equal volume ≠ equal surface area; (3) a geometric reason (e.g., 'the cylinder's tall straight sides create more lateral area than the cone's slanted surface converging to a point'). Learners missing indicator (2) need targeted follow-up. Learners missing indicator (3) are ready for extension: 'Can you find a frustum dimension set where the frustum really does use the most metal?' |

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| | | learners who held the minority view in the prediction phase. | | |
| **Phase 4 — Driving Question Board (DQB) Finalisation**<br>📹 **VIDEO:** Writing proportional equations from tables<br>Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | The class gathers around the DQB that has been built across Lessons 1–7. Each question card is read aloud by a volunteer. The class votes: GREEN (fully answered with evidence), AMBER (partially answered — we know more but questions remain), or RED (new question opened up by our learning). Learners use sticky dots (green, amber, red) to mark each card. For every GREEN card, a nominated learner writes a one-sentence 'Answer Summary' on a new card and pins it beside the question. For AMBER or RED cards, learners discuss briefly what further investigation would be needed. The driving question itself — 'How do we calculate exactly how much material is needed to cover or construct any solid shape — and why does the shape of a container matter even when the volume stays the same?' — is read aloud as the final DQB action, and learners write a collective two-sentence answer on a large card that is pinned at the centre of the board. | Lead the DQB session from the front. Read each question card clearly and slowly. Say: 'Give me a thumbs up for green, sideways for amber, down for red.' Wait 5 seconds after each card for learners to decide. For questions rated GREEN, say: 'Who can give us the one-sentence answer?' Cold-call using name sticks — wait 10 seconds. Write the summary on the card yourself (or have the learner do it) and pin it prominently. For AMBER cards, say: 'What would we need to learn or measure to fully answer this?' — wait 12 seconds, accept 2 responses. For the driving question card at the end, say: 'This is the question that started our entire journey in this unit. Turn to your partner and agree on two sentences that answer it. You have 90 seconds.' Cold-call two pairs to share, then synthesise a class answer on the board card. Express genuine affirmation: 'You walked into this unit not knowing how to calculate the surface area of a frustum. Look at what you can do now.' | Driving Question Board Finalisation (NGSS Practice 1 — Asking Questions; Science and Engineering Practice 8 — Obtaining, Evaluating, and Communicating Information): Revisiting every DQB card makes the trajectory of learning visible and gives learners a metacognitive 'map' of their conceptual growth across the unit. The traffic-light voting ritual is quick, inclusive, and surfaces remaining gaps before the unit closes. | Observe which question cards are marked AMBER or RED — these signal persistent conceptual gaps or productive extensions for motivated learners. If more than 30% of cards about frustum surface area are AMBER, note this for revision before the end-of-strand assessment. Observable indicator of unit-level mastery: learner can spontaneously answer the driving question in two coherent sentences without prompting. |
| **Phase 5 — Model Building: Portfolio Task and Cumulative Model Revision**<br>📹 **VIDEO:** Writing proportional equations from tables | Learners complete two parallel tasks. TASK A (Portfolio Task — individual, 15 minutes): Each learner selects ONE real-life Kenyan object from a provided list — (i) a conical grain store (granary) in a Kisii homestead, (ii) a frustum-shaped Ndovu water tank common in Kisumu, (iii) a spherical elevated water tank in Nairobi, (iv) a square-pyramid-roofed community hall in Eldoret, or (v) their own chosen Kenyan | Distribute Portfolio Record Sheets. Say: 'For the next 15 minutes, every one of you is a mathematician working independently. Choose your Kenyan object, write the formula, substitute, and calculate. Show every line of working — a correct answer with no working earns half marks.' Circulate and offer targeted prompts only — avoid giving answers. For the grain-store cone: 'What is the slant height? Do | Portfolio Task + Cumulative Model Revision (NGSS Practice 2 — Developing and Using Models; NGSS Practice 8 — Obtaining, Evaluating, and Communicating Information): The portfolio task operationalises transfer — learners apply formulae to a new, self-chosen Kenyan context, demonstrating competency beyond the taught examples. The cumulative model revision makes conceptual growth tangible and | Collect Portfolio Record Sheets at the end of the lesson. Assess using the KICD rubric indicators: (i) correct solid type identification, (ii) correct formula stated, (iii) correct substitution with units, (iv) correct final answer with units, (v) meaningful real-world interpretation sentence. Learners meeting all five indicators 'Exceed Expectations'. Learners completing (i)–(iv) correctly 'Meet Expectations'. Also scan |

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| Source: Khan Academy (English - US curriculum)<br>🔍 Search ARES for similar videos<br><br>📖 **HTML:** Surface Area and Nets (Flexbook)<br>Source: CK-12<br>🔍 Search ARES for similar readings | object with teacher approval. They are given a 'Portfolio Record Sheet' with estimated/given dimensions and must: label the solid type, write the applicable formula, substitute the dimensions, and calculate the total surface area. They also write one sentence explaining what the surface area represents in the real-world context (e.g., 'This is the total area of sheet metal needed to fabricate the Ndovu water tank, which determines the cost of materials at the hardware shop.'). TASK B (Cumulative Model Revision — pairs, 10 minutes): Pairs retrieve their cumulative surface area model (the annotated diagram/poster built progressively across Lessons 1–7). They add the final solids or annotations not yet included, ensure all five solid types (cone, pyramid, sphere, hemisphere, frustum) are labelled with formulae, and draw arrows connecting each solid back to the anchoring phenomenon buckets. The completed model is signed by both learners. | you need to calculate it from the radius and perpendicular height using Pythagoras?' Wait 10 seconds before offering the hint. After 15 minutes, say: 'Now retrieve your cumulative model with your partner. You have 10 minutes to make it complete — every solid, every formula, every connection back to our three buckets on Kirinyaga Road.' At the 5-minute mark, say: 'Check: is the frustum formula on your model? Is there an arrow from the frustum formula to the Kirinyaga Road bucket?' Close the lesson by asking 3 pairs to hold up their completed models for the class to see. Say: 'This model is the evidence of everything you have learned in Sub-Strand 2.6. It belongs to you — keep it as a study tool.' | spatial, reinforcing the connected structure of the sub-strand. Together, these tasks constitute the summative evidence artefact for the unit. | cumulative models: look for all five solid formulae present, correct notation (e.g., l for slant height, r and R for frustum radii), and at least one explicit connection drawn back to the anchoring phenomenon buckets. |

## D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the Evidence Court in Phase 3 produce genuine surprise when learners found that the open cylinder — not the frustum — used the most metal under these specific equal-volume dimensions? If most groups had predicted the frustum and were surprised, the phenomenon successfully generated productive cognitive dissonance. If the result was accepted without curiosity, consider whether learners understood WHY — re-examine the Final Explanation paragraphs for depth of geometric reasoning. (2) Were the DQB GREEN cards answered with precision, or were the one-sentence summaries vague? Vague summaries suggest learners can execute calculations but cannot articulate conceptual understanding — plan a verbal explanation task for the revision lesson before the end-of-strand assessment. (3) Did learners select diverse Kenyan objects for the portfolio task, or did most gravitate to the same solid type? Lack of diversity may indicate a preference for the 'easiest' formula rather than genuine connection to real-life contexts. (4) How complete and accurate were the cumulative models? Missing formulae or absent connections to the phenomenon suggest the progressive model-building across the unit did not consolidate effectively — consider introducing a model-review checkpoint mid-unit in future iterations. (5) Were all learners able to contribute meaningfully to group presentations, or did one or two learners dominate? Adjust group composition or role assignments in future consolidation lessons to ensure equitable participation. Note any learners who still conflate volume and surface area at the end of this final lesson — these learners require individual follow-up before the Sub-Strand 2.7 (Volume and Capacity) assessment, as the confusion will compound.

| E. SUMMARY TABLE PROMPT (pre-filled example for this lesson) | |
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| **What did I observe?** | Learners calculated the surface areas of three buckets — cylindrical, conical, and frustum-shaped — all holding exactly 10 litres of water, using dimensions provided on the Bucket Data Sheet. They displayed their full working on sugar-paper sheets and presented findings to the class. Learners also observed, via the DQB review, the full set of question cards built across eight lessons. |
| **What did I learn?** | Equal volumes do NOT guarantee equal surface areas — the shape of a solid determines how much material is needed to construct it, independently of its capacity. The frustum's formula $\pi(R+r)l + \pi r^2$ accounts for two different circular radii and a slant height, making it more complex than the cone or cylinder formula. Under the specific equal-volume dimensions used in this lesson, the open cylinder required the greatest surface area of sheet metal. Each of the five solid types studied (cone, pyramid, sphere, hemisphere, frustum) has a unique surface area formula derived from its net. |
| **How does this explain the phenomenon?** | The shopkeeper's claim that the frustum uses the most metal is not universally correct — it depends on the specific dimensions chosen. For our 10-litre buckets, the open cylinder used more sheet metal than the frustum. The shape of a container matters for material cost because surface area and volume are related by different geometric formulae: changing the shape while keeping the volume fixed changes the surface area. This is why engineers and manufacturers choose specific solid shapes — to minimise material cost, maximise strength, or meet practical requirements like stackability. In Kenya, frustum-shaped buckets are common despite potentially higher material costs because their tapered shape makes them easier to stack and carry, demonstrating that mathematical optimisation in real life must balance multiple competing factors. |

| DIFFERENTIATION AND INCLUSION | | |
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| **Learner Need** | **Support Strategies** | **Extension Strategies** |
| English Language Learners | Provide bilingual glossaries; allow use of first language for initial model drawing. | Write extended explanations of observations; lead group discussions in English. |
| Students with learning difficulties | Provide structured note frames; pair with a supportive partner during investigations. | Mentor peers; create additional visual models to explain concepts. |
| Gifted and advanced learners | Access to additional reading materials; challenge questions during DQB. | Lead model-building discussions; research and present extension topics to class. |